A Dirichlet Pólya Inequality for Three-Dimensional Spherical Shells

Purely analytic main proof manuscript

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Scope and status. This manuscript proves a theorem for the class of genuine three-dimensional spherical shells. It does not claim the general Pólya conjecture, literature novelty, or priority. All shell-specific analytic reductions, regional arguments, seam assignments, exact constants, and all finite exact-rational decisions are stated in this paper and its accompanying purely analytic supplement. Executable replay sources and interval certificates are retained only as optional regression checks and are not premises of the proof. The imported background and analytic results comprise explicitly identified standard Sturm–Liouville and Bessel facts and published phase, Bessel-zero, and one-dimensional floor-sum inequalities, each quoted with its precise scope.

Abstract

For $0 < r < R$, let

$$A_{r,R} = \{x \in \mathbb{R}^3 : r < |x| < R\}.$$

We prove, with the strict Dirichlet counting convention, that

$$N_D(A_{r,R}, \Lambda) \leq L_3 |A_{r,R}| \Lambda^{3/2}, \qquad L_3 = \frac{1}{6\pi^2}, \qquad \Lambda \geq 0.$$

The proof is purely analytic. It separates variables, converts the radial spectrum into a strict multiplicity-weighted phase count, and combines analytic endpoint estimates, finite angular and radial mode inventories, hand-checkable exact-rational ledgers, a retained-remainder identity, and an upward-monotone scalar estimate beyond the final staircase. No floating-point or interval-arithmetic conclusion is used as a proof premise. We also give an elementary proof that no genuine spherical shell tiles $\mathbb{R}^3$ by congruent rigid-motion copies with disjoint interiors, under either exact or almost-everywhere coverage.

Contents

1 Main results and conventions

The eigenvalues of the Dirichlet Laplacian on a bounded domain Ω are listed with multiplicity as

$$0 < \lambda_1^D(\Omega) \le \lambda_2^D(\Omega) \le \cdots .$$

Throughout we use the strict counting function

$$N_D(\Omega, \Lambda) := \#\{j : \lambda_j^D(\Omega) < \Lambda\}. \tag{1.1}$$

This convention matters at every spectral wall. We write $\mathbb{N} = \{1, 2, 3, \ldots\}$ and $\mathbb{N}_0 = \{0, 1, 2, \ldots\}$.

Theorem 1.1 (Spectral inequality)**.** *For every* $0 < r < R$ *and every* $\Lambda \ge 0$*,*

$$\boxed{N_D(A_{r,R}, \Lambda) \le \frac{2}{9\pi}(R^3 - r^3)\Lambda^{3/2} = L_3|A_{r,R}|\Lambda^{3/2}, \qquad L_3 = \frac{1}{6\pi^2}.} \tag{1.2}$$

Theorem 1.2 (Non-tiling)**.** *For every* $0 < r < R$*, no family of congruent rigid-motion copies of* $A_{r,R}$ *has pairwise-disjoint interiors and covers* $\mathbb{R}^3$ *exactly or up to a three-dimensional Lebesgue-null set. The same conclusion holds when coverage is formulated using closed shells.*

The two theorems concern literally the same complete shell family. Theorem 1.2 is logically independent of the spectral proof.

2 Separated spectrum, phase reduction, and the small-hole endpoint

Throughout this section

$$A_{\rho,1} = \{x \in \mathbb{R}^3 : \rho < |x| < 1\}, \qquad 0 < \rho < 1,$$

and the Dirichlet counting function is strict:

$$N_D(A_{\rho,1}, K^2) := \#\{j : \lambda_j^D(A_{\rho,1}) < K^2\}.$$

For $x > 0$ we use the strict positive-integer count

$$[x]_< := \#\{n \in \mathbb{N} : n < x\} = \max\{0, \lceil x \rceil - 1\}. \tag{2.1}$$

Thus $[m]_< = m - 1$ for $m \in \mathbb{N}$. This convention is retained at every eigenfrequency, phase level, and half-integer angular wall.

2.1 The exact separated spectrum

Let $H_\rho = -\Delta^D_{A_{\rho,1}}$ be the Friedrichs operator associated with

$$\mathfrak{q}[f] = \int_{A_{\rho,1}} |\nabla f|^2 \, dx, \qquad D(\mathfrak{q}) = H_0^1(A_{\rho,1}).$$

Choose an orthonormal spherical-harmonic basis satisfying

$$-\Delta_{\mathbb{S}^2} Y_{\ell m} = \ell(\ell + 1) Y_{\ell m}, \qquad \ell \in \mathbb{N}_0, \quad -\ell \le m \le \ell.$$

Proposition 2.1 (Unitary decomposition and completeness)**.** *For*

$$L_\ell = -\frac{d^2}{dr^2} + \frac{\ell(\ell+1)}{r^2}, \qquad D(L_\ell) = H^2(\rho,1) \cap H_0^1(\rho,1),$$

one has the unitary equivalence

$$H_\rho \simeq \bigoplus_{\ell=0}^{\infty} \bigoplus_{m=-\ell}^{\ell} L_\ell. \tag{2.2}$$

The exact form domain on the right is

$$\left\{ (u_{\ell m}) : \begin{array}{l} u_{\ell m} \in H_0^1(\rho,1), \\ \sum_{\ell,m} \Big(\|u_{\ell m}\|_2^2 + \mathfrak{a}_\ell[u_{\ell m}] \Big) < \infty \end{array} \right\}, \tag{2.3}$$

where

$$\mathfrak{a}_\ell[u] = \int_\rho^1 \left(|u'|^2 + \frac{\ell(\ell+1)}{r^2}|u|^2 \right) dr.$$

The exact operator domain is

$$\left\{ (u_{\ell m}) : \begin{array}{l} u_{\ell m} \in D(L_\ell), \\ \sum_{\ell,m} \Big(\|u_{\ell m}\|_2^2 + \|L_\ell u_{\ell m}\|_2^2 \Big) < \infty \end{array} \right\}. \tag{2.4}$$

Each L_ℓ has positive, simple eigenvalues tending to infinity and a complete orthonormal eigenbasis. The full operator has compact resolvent.

Proof. Write $x = r\omega$, so $dx = r^2\, dr\, d\omega$, and set

$$R_{\ell m}(r) = \int_{\mathbb{S}^2} f(r,\omega)\overline{Y_{\ell m}(\omega)}\, d\omega, \qquad u_{\ell m}(r) = rR_{\ell m}(r).$$

Fubini, Tonelli, and Parseval on $\mathbb{S}^2$ give

$$\|f\|_{L^2(A_{\rho,1})}^2 = \sum_{\ell,m} \int_\rho^1 |R_{\ell m}(r)|^2 r^2\, dr = \sum_{\ell,m} \int_\rho^1 |u_{\ell m}(r)|^2\, dr.$$

Conversely, if $(u_{\ell m})$ is square summable, the finite sums

$$f_L(r,\omega) = \frac{1}{r} \sum_{\ell \le L} \sum_{m=-\ell}^{\ell} u_{\ell m}(r) Y_{\ell m}(\omega)$$

are Cauchy in $L^2(A_{\rho,1})$, and their limit maps back to the given family. Hence the coefficient map is unitary onto the Hilbert direct sum. Because $r, 1/r \in W^{1,\infty}(\rho,1)$, multiplication by either function is bounded on $H^1(\rho,1)$ and preserves the two zero endpoint traces.

For a finite smooth harmonic expansion, polar coordinates and angular orthogonality yield

$$\mathfrak{q}[f] = \sum_{\ell,m} \int_\rho^1 \Big(r^2|R_{\ell m}'|^2 + \ell(\ell+1)|R_{\ell m}|^2 \Big)\, dr.$$

If $u = rR$, then

$$r^2|R'|^2 = \left| u' - \frac{u}{r} \right|^2 = |u'|^2 - \frac{(|u|^2)'}{r} + \frac{|u|^2}{r^2}.$$

Since

$$\frac{(|u|^2)'}{r} = \left(\frac{|u|^2}{r}\right)' + \frac{|u|^2}{r^2}$$

and $u(\rho) = u(1) = 0$, integration gives

$$\int_\rho^1 r^2|R'|^2\,dr = \int_\rho^1 |u'|^2\,dr.$$

The angular term becomes $\ell(\ell+1)|u|^2/r^2$. Closing finite harmonic expansions in the form norm proves (2.3): one inclusion follows from componentwise form convergence and Fatou's lemma, while the reverse inclusion follows by first truncating the square-summable family in (ℓ, m) and then approximating its finitely many components by $C_c^\infty(\rho, 1)$. The uniqueness theorem for operators associated with closed semibounded forms gives (2.2).

For fixed ℓ, the potential $\ell(\ell+1)/r^2$ is continuous and bounded on $[\rho, 1]$. The regular one-dimensional Dirichlet form therefore has associated operator L_ℓ with the displayed $H^2 \cap H_0^1$ domain. The standard domain formula for an orthogonal operator sum gives (2.4).

The embedding $H_0^1(\rho, 1) \hookrightarrow L^2(\rho, 1)$ is compact, so every block has compact resolvent. To pass to the infinite direct sum, note that $r \leq 1$ implies

$$L_\ell \geq \ell(\ell+1), \qquad \|(L_\ell + 1)^{-1}\| \leq \frac{1}{1+\ell(\ell+1)}.$$

The resolvent of the full operator is consequently the operator-norm limit of its finite-ℓ truncations, each of which is compact. Hence the full resolvent is compact.

Regular Dirichlet Sturm–Liouville theory gives a complete sequence of simple eigenvalues in every block. They are strictly positive because

$$\mathfrak{a}_\ell[u] \geq \int_\rho^1 |u'|^2\,dr \geq \frac{\pi^2}{(1-\rho)^2}\|u\|_2^2.$$

Simplicity also follows directly: two solutions of the same second-order equation that vanish at $r = \rho$ have proportional initial derivatives and hence are proportional by uniqueness. If $u_{\ell,n}$ is a complete radial eigenbasis, then

$$\frac{u_{\ell,n}(r)}{r} Y_{\ell m}(\omega)$$

over all (ℓ, m, n) is exactly the inverse image of a complete orthonormal coordinate basis of the Hilbert direct sum. This proves global completeness. □

Proposition 2.2 (Bessel determinant and strict phase enumeration)**.** *Put* $\nu = \ell + \frac{1}{2}$. *The positive eigenfrequencies in the* ℓ*-th radial block are precisely the positive zeros of*

$$D_{\nu,\rho}(k) := J_\nu(\rho k)Y_\nu(k) - Y_\nu(\rho k)J_\nu(k). \tag{2.5}$$

Every such zero is simple. With the continuous phase branch

$$J_\nu(x) + iY_\nu(x) = M_\nu(x)e^{i\theta_\nu(x)}, \qquad M_\nu(x) > 0, \qquad \theta_\nu(0+) = -\frac{\pi}{2}, \tag{2.6}$$

and

$$\Psi_{\nu,\rho}(K) = \theta_\nu(K) - \theta_\nu(\rho K), \qquad \gamma_{\rho,K}(\nu) = \frac{\Psi_{\nu,\rho}(K)}{\pi},$$

the exact strict count is

$$N_D(A_{\rho,1}, K^2) = \sum_{\ell=0}^{\infty}(2\ell+1)[\gamma_{\rho,K}(\ell + \tfrac{1}{2})]_<. \tag{2.7}$$

The sum is finite. If K *is a radial eigenfrequency in one or several angular channels, the root at* K *is excluded in each such channel and the excluded angular multiplicities add.*

Proof. For $k > 0$, the physical radial equation has the general solution

$$R(r) = r^{-1/2}\big(c_1 J_\nu(kr) + c_2 Y_\nu(kr)\big).$$

The two Dirichlet conditions admit a nonzero coefficient vector exactly when

$$\det\begin{pmatrix} J_\nu(\rho k) & Y_\nu(\rho k) \\ J_\nu(k) & Y_\nu(k) \end{pmatrix} = D_{\nu,\rho}(k) = 0.$$

The Wronskian

$$J_\nu(x)Y_\nu'(x) - J_\nu'(x)Y_\nu(x) = \frac{2}{\pi x} \tag{2.8}$$

shows that the two entries in either row cannot vanish simultaneously. Thus the determinant criterion remains exact even if the J_ν values, or the Y_ν values, happen to vanish at both endpoints.

For root simplicity define the left-normalized solution

$$s_\ell(r,k) = \frac{\pi\sqrt{\rho}}{2}\sqrt{r}\left[J_\nu(\rho k)Y_\nu(kr) - Y_\nu(\rho k)J_\nu(kr)\right]. \tag{2.9}$$

Equation (2.8) gives

$$s_\ell(\rho,k) = 0, \qquad \partial_r s_\ell(\rho,k) = 1, \qquad s_\ell(1,k) = \frac{\pi\sqrt{\rho}}{2}D_{\nu,\rho}(k).$$

Let $v = \partial_k s_\ell$. Differentiating $(L_\ell - k^2)s_\ell = 0$ gives

$$(L_\ell - k^2)v = 2ks_\ell, \qquad v(\rho,k) = \partial_r v(\rho,k) = 0.$$

At a positive determinant root, Lagrange's identity yields

$$\partial_k s_\ell(1,k)\,\partial_r s_\ell(1,k) = 2k\int_\rho^1 s_\ell(r,k)^2\,dr > 0. \tag{2.10}$$

Thus $\partial_k s_\ell(1,k) \neq 0$, and the zero of the determinant is simple.

At zero energy the left-normalized solution is

$$s_\ell(r,0) = \frac{\rho^{-\ell}r^{\ell+1} - \rho^{\ell+1}r^{-\ell}}{2\ell+1},$$

so

$$s_\ell(1,0) = \frac{\rho^{-\ell} - \rho^{\ell+1}}{2\ell+1} > 0.$$

Equivalently, the standard small-argument expansions give

$$\lim_{k\downarrow 0} D_{\nu,\rho}(k) = \frac{\rho^{-\nu} - \rho^{\nu}}{\pi\nu} > 0. \tag{2.11}$$

Hence the limiting phase level zero is not an eigenfrequency.

From (2.6),

$$J_\nu = M_\nu\cos\theta_\nu, \qquad Y_\nu = M_\nu\sin\theta_\nu,$$

and direct substitution gives the sign-sensitive factorization

$$D_{\nu,\rho}(k) = M_\nu(\rho k)M_\nu(k)\sin\Psi_{\nu,\rho}(k). \tag{2.12}$$

The Wronskian also gives

$$\theta_\nu'(x) = \frac{2}{\pi x M_\nu(x)^2} > 0.$$

Nicholson's integral representation

$$M_\nu(x)^2 = \frac{8}{\pi^2}\int_0^\infty \cosh(2\nu t) K_0(2x\sinh t)\, dt$$

and the strict decrease of K_0 imply that $M_\nu(x)^2$ is strictly decreasing in x. Therefore

$$\Psi_{\nu,\rho}'(k) = \frac{2}{\pi k}\left(M_\nu(k)^{-2} - M_\nu(\rho k)^{-2}\right) > 0. \tag{2.13}$$

The branch condition gives $\Psi_{\nu,\rho}(0+) = 0$, while the standard large-argument Bessel expansion gives

$$\Psi_{\nu,\rho}(k) = (1-\rho)k + O(k^{-1}) \longrightarrow \infty$$

for fixed ν and ρ. Hence the positive determinant roots are uniquely indexed by

$$\Psi_{\ell+1/2,\rho}(k_{\ell,n}) = n\pi, \qquad n = 1, 2, \ldots.$$

Strict monotonicity gives

$$k_{\ell,n} < K \quad \Longleftrightarrow \quad n\pi < \Psi_{\ell+1/2,\rho}(K).$$

The direct-sum decomposition supplies angular multiplicity $2\ell+1$. If the same numerical eigenvalue occurs at distinct values of ℓ, the corresponding spherical-harmonic sectors are orthogonal and their multiplicities add. Finally, $\lambda_{\ell,n} \geq \ell(\ell+1)$, so only finitely many angular channels can contribute below K^2. These observations prove (2.7), including every strict spectral wall. □

The only standard inputs in the preceding proof are completeness of the spherical harmonics, regular one-dimensional Dirichlet Sturm–Liouville theory, the Bessel equation, the Wronskian, Nicholson's representation, and the usual small- and large-argument Bessel formulas; see [5].

2.2 Uniform phase enclosures and the strict lattice proxy

For $a > 0$, define

$$G_a(z) = \begin{cases} \dfrac{1}{\pi}\left(\sqrt{a^2 - z^2} - z\arccos\dfrac{z}{a}\right), & 0 \leq z \leq a, \\ 0, & z > a. \end{cases} \tag{2.14}$$

For $0 \leq z < a$, put

$$H_a(z) = \frac{3a^2 + 2z^2}{24\pi(a^2 - z^2)^{3/2}},$$

and define

$$\mathcal{F}_a(z) = \begin{cases} \max\{G_a(z) - H_a(z), -1/4\}, & 0 \leq z < a, \\ -1/4, & z \geq a. \end{cases} \tag{2.15}$$

We use the following published Bessel-phase result, stated here with every hypothesis needed in the sequel.

Lemma 2.3 (External uniform Bessel-phase enclosure)**.** *With the branch (2.6), for every $a > 0$ and real $z \geq 0$,*

$$\mathcal{F}_a(z) - \frac{1}{4} < \frac{\theta_z(a)}{\pi} < G_a(z) - \frac{1}{4}. \tag{2.16}$$

If $0 < \mu < K$, $0 \leq z \leq K$, and

$$\Gamma_{K,\mu}(z) := \frac{\theta_z(K) - \theta_z(\mu)}{\pi},$$

then

$$0 < \Gamma_{K,\mu}(z) < G_K(z) - \mathcal{F}_\mu(z) \leq G_K(z) + \frac{1}{4}. \tag{2.17}$$

If $0 \leq z \leq \mu$, including $z = \mu$, then also

$$G_K(z) - G_\mu(z) < \Gamma_{K,\mu}(z) < G_K(z) - G_\mu(z) + \frac{1}{4}. \tag{2.18}$$

No value of the divergent expression $H_\mu(\mu)$ is used in the last statement.

Lemma 2.3 is Theorem 5.1 and Lemma 5.2 of [3], based on the real-order phase enclosures of [2]. The upper bound in (2.17) follows immediately by subtracting the lower bound for $\theta_z(\mu)$ from the upper bound for $\theta_z(K)$ in (2.16); the correlated two-sided estimate (2.18) is the sharper part of the annulus result. Both theorems are stated for real order, so the half-integers $z = \ell + \frac{1}{2}$ require no interpolation.

Fix now $K > 0$, set $\mu = \rho K$, and define

$$A_{\rho,K}(z) = G_K(z) - G_\mu(z). \tag{2.19}$$

For $0 \leq z \leq K$, let

$$s_\mu(z) = \begin{cases} \min\{H_\mu(z), 1/4\}, & 0 \leq z < \mu, \\ 1/4, & \mu \leq z \leq K, \end{cases} \qquad U_{\rho,K}(z) = A_{\rho,K}(z) + s_\mu(z). \tag{2.20}$$

The second line, rather than H_μ, is always used at $z = \mu$.

Proposition 2.4 (Strict phase-to-lattice reduction)**.** *For $0 < \rho < 1$ and $K > 0$,*

$$\gamma_{\rho,K}(z) < U_{\rho,K}(z) \leq A_{\rho,K}(z) + \frac{1}{4}, \qquad 0 \leq z \leq K. \tag{2.21}$$

Because every G_a is zero-extended, $A_{\rho,K}$ is the piecewise action $G_K - G_{\rho K}$ below the inner interface and G_K at and above it. Consequently,

$$N_D(A_{\rho,1}, K^2) \leq \sum_{\nu_\ell < K} 2\nu_\ell [U_{\rho,K}(\nu_\ell)]_< \tag{2.22}$$

$$\leq S_{\rho,K} := \sum_{\nu_\ell < K} 2\nu_\ell \left\lfloor A_{\rho,K}(\nu_\ell) + \frac{1}{4} \right\rfloor, \qquad \nu_\ell = \ell + \frac{1}{2}. \tag{2.23}$$

The cutoff is strict: an order $\nu_\ell = K$ is inactive. At a true phase wall the radial count is one less than the ordinary floor; at a proxy wall the ordinary floor in (2.23) is retained. Moreover,

$$\int_0^\infty 2z A_{\rho,K}(z)\, dz = \frac{2}{9\pi}(1 - \rho^3) K^3. \tag{2.24}$$

Proof. For $0 \le z < \mu$, the first upper bound in (2.17) reads

$$\gamma_{\rho,K}(z) < A_{\rho,K}(z) + G_\mu(z) - \mathcal{F}_\mu(z).$$

From (2.15),

$$G_\mu(z) - \mathcal{F}_\mu(z) = \min\{H_\mu(z), G_\mu(z) + 1/4\}.$$

The upper half of (2.18) also gives $\gamma_{\rho,K}(z) < A_{\rho,K}(z) + 1/4$. Taking the better of these two strict upper bounds, and using $G_\mu(z) + 1/4 \ge 1/4$, gives

$$\gamma_{\rho,K}(z) < A_{\rho,K}(z) + \min\{H_\mu(z), 1/4\} = U_{\rho,K}(z).$$

For $\mu \le z \le K$, one has $G_\mu(z) = 0$, and (2.17) gives

$$\gamma_{\rho,K}(z) < G_K(z) + 1/4 = U_{\rho,K}(z).$$

Orders $z \ge K$ do not contribute. Indeed, (2.16) gives

$$\frac{\theta_z(K)}{\pi} < -\frac{1}{4}, \qquad \frac{\theta_z(\mu)}{\pi} > -\frac{1}{2},$$

because $G_K(z) = 0$ and $\mathcal{F}_\mu(z) = -1/4$. Together with phase monotonicity this yields

$$0 < \gamma_{\rho,K}(z) < \frac{1}{4} < 1,$$

so there is no positive phase level below K. This includes $z = K$.

For positive $x < y$, every positive integer strictly below x is also strictly below y, so $[x]_< \le [y]_< \le \lfloor y \rfloor$. Applying this observation to the exact phase count (2.7), and then using $s_\mu \le 1/4$, proves (2.22)– (2.23) without changing either wall convention.

Finally, scaling $z = ax$ gives

$$\int_0^a 2zG_a(z)\,dz = \frac{2a^3}{\pi}\left(\int_0^1 x\sqrt{1-x^2}\,dx - \int_0^1 x^2 \arccos x\,dx\right).$$

The two elementary integrals are $1/3$ and $2/9$, respectively; hence the last expression is $2a^3/(9\pi)$. Subtracting the result with $a = \mu = \rho K$ from that with $a = K$ proves (2.24). □

The necessity of retaining integerization can be displayed exactly. Put

$$m_K = \max\{0, \lceil K - \tfrac{1}{2} \rceil\}, \qquad r_\mu = \max\{0, \lceil \mu - \tfrac{1}{2} \rceil\}.$$

Thus the active orders are $\nu_0, \dots, \nu_{m_K-1}$, and precisely the first r_μ of them lie strictly below μ. Define

$$E_0(a) = \sum_{\nu_\ell < a} 2\nu_\ell G_a(\nu_\ell) - \frac{2a^3}{9\pi}, \qquad E_{\mathrm{mp}}(\rho, K) = E_0(K) - E_0(\mu),$$

$$E_{\mathrm{ph}} = \sum_{\ell=0}^{r_\mu - 1} (2\ell+1)\min\{H_\mu(\nu_\ell), 1/4\} + \frac{m_K^2 - r_\mu^2}{4}.$$

For $x > 0$, define the strict remainder

$$\langle x \rangle_< := x - [x]_< \in (0, 1],$$

so $\langle m\rangle_< = 1$ at positive integer m, and set

$$E_{\mathrm{frac}} = \sum_{\nu_\ell<K} 2\nu_\ell \langle U_{\rho,K}(\nu_\ell)\rangle_<.$$

Expanding $U = A + s_\mu$ term by term and using (2.24) gives the exact identity

$$\sum_{\nu_\ell<K} 2\nu_\ell [U_{\rho,K}(\nu_\ell)]_< - \frac{2}{9\pi}(1-\rho^3)K^3 = E_{\mathrm{mp}} + E_{\mathrm{ph}} - E_{\mathrm{frac}}. \tag{2.25}$$

In particular, the blanket quarter-slack cost is exactly

$$\frac{1}{4}\sum_{\ell=0}^{m_K-1}(2\ell+1) = \frac{m_K^2}{4};$$

it cannot be discarded merely because it is of lower polynomial order for fixed ρ.

2.3 Multiplicity reduction to shifted one-dimensional tails

We record the two external floor-sum inequalities used in the small-hole argument. For integers $a < b$, write

$$\mathbf{T}(h;a,b) := \frac{1}{2}\lfloor h(a)\rfloor + \sum_{j=a+1}^{b-1}\lfloor h(j)\rfloor + \frac{1}{2}\lfloor h(b)\rfloor. \tag{2.26}$$

Lemma 2.5 (External trapezoidal floor bounds)**.** *The following statements hold.*

1. *If h is nonnegative and concave on an integer interval $[a,b]$, then*

$$\mathbf{T}(h;a,b) \le \int_a^b h(t)\,dt. \tag{2.27}$$

2. *Suppose h is nonnegative, decreasing, concave, and c-Lipschitz on an integer interval $[a,b]$, where $0<c<1$. If an integer $p \in [a,b)$ is the last point of the first ordinary-floor shelf, so that*

$$\lfloor h(a)\rfloor = \lfloor h(p)\rfloor > \lfloor h(p+1)\rfloor,$$

then

$$\mathbf{T}(h;a,b) \le \int_a^b h(t)\,dt - \frac{1-c}{2}(b-p). \tag{2.28}$$

If no such drop occurs, set $p=b$; the basic bound in item 1 gives the same displayed formula with zero correction. This last no-drop case is a consequence of the basic proposition, not an invocation of the improved theorem at $p=b$.

3. *If g is nonnegative, decreasing, convex, and $1/2$-Lipschitz on an integer interval $[a,b]$, if $g(b)=0$, and if g is $1/3$-Lipschitz on $[t_*,b]$ for some $t_* \in [a,b]$, then*

$$\mathbf{T}(g+1/4;a,b) \le \int_a^b g(t)\,dt - \frac{1}{4}\lfloor g(t_*)\rfloor. \tag{2.29}$$

4. *If g is nonnegative, decreasing, convex, and $1/2$-Lipschitz on $[0,b]$, where $b>0$ is arbitrary and $g(b)=0$, then*

$$\left\lfloor g(0)+\frac{1}{4}\right\rfloor + 2\sum_{j=1}^{\lfloor b\rfloor}\left\lfloor g(j)+\frac{1}{4}\right\rfloor \le 2\int_0^b g(t)\,dt. \tag{2.30}$$

The first assertion is Proposition 3.1 of [3]; the first-shelf improvement is its Theorem 1.4; and the third assertion is its Theorem 3.4. The fourth is the translated convex shifted-floor theorem of [1]. These published one-dimensional inequalities are the only non-special-function external inputs in what follows.

For $r \in \mathbb{N}_0$, put $x_r = r + \frac{1}{2}$ and define the coarse shifted tail

$$\mathcal{T}_r = \left\lfloor A_{\rho,K}(x_r) + \frac{1}{4} \right\rfloor + 2 \sum_{j \geq 1} \left\lfloor A_{\rho,K}(x_r + j) + \frac{1}{4} \right\rfloor. \tag{2.31}$$

All but finitely many terms vanish.

Lemma 2.6 (Weighted-lattice scaffold)**.** *One has*

$$S_{\rho,K} = \sum_{r \geq 0} \mathcal{T}_r. \tag{2.32}$$

If, for every r with $x_r < K$,

$$\mathcal{T}_r \leq 2 \int_{x_r}^{K} A_{\rho,K}(z)\, dz, \tag{2.33}$$

then

$$S_{\rho,K} \leq \frac{2}{9\pi}(1 - \rho^3)K^3. \tag{2.34}$$

Proof. For any finitely supported nonnegative integer sequence q_ℓ,

$$\sum_{\ell \geq 0} (2\ell + 1) q_\ell = \sum_{r \geq 0} \left(q_r + 2 \sum_{\ell > r} q_\ell \right).$$

Indeed, q_ℓ appears once with coefficient one when $r = \ell$ and appears ℓ times with coefficient two when $0 \leq r < \ell$. Taking

$$q_\ell = \left\lfloor A_{\rho,K}(\ell + \tfrac{1}{2}) + \frac{1}{4} \right\rfloor$$

proves (2.32); samples at or beyond K are zero.

Assume (2.33). Tonelli's theorem for the finite nonnegative sum gives

$$S_{\rho,K} \leq 2 \int_0^K f_3(z) A_{\rho,K}(z)\, dz, \qquad f_3(z) = \#\{r \in \mathbb{N}_0 : r + \tfrac{1}{2} \leq z\}.$$

Let $F_3(z) = \int_0^z f_3(t)\, dt$. If $z = m + \frac{1}{2} + s$, where $m \in \mathbb{N}_0$ and $0 \leq s < 1$, then direct summation gives

$$\frac{z^2}{2} - F_3(z) = \frac{1}{2}\left(s - \frac{1}{2}\right)^2 \geq 0; \tag{2.35}$$

for $0 \leq z < 1/2$ the same inequality is immediate because $F_3(z) = 0$. The action $A_{\rho,K}$ is absolutely continuous, decreasing, and vanishes at K. Indeed, away from the inner interface,

$$A'_{\rho,K}(z) = \begin{cases} -\dfrac{1}{\pi}\left(\arccos\dfrac{z}{K} - \arccos\dfrac{z}{\mu}\right) < 0, & 0 < z < \mu, \\ -\dfrac{1}{\pi}\arccos\dfrac{z}{K} < 0, & \mu < z < K, \end{cases}$$

and the two formulas have the same limiting value at $z = \mu$. Integration by parts and (2.35) therefore give

$$2\int_0^K f_3 A_{\rho,K} = -2\int_0^K F_3 A'_{\rho,K} \leq -\int_0^K z^2 A'_{\rho,K}(z)\, dz = \int_0^K 2z A_{\rho,K}(z)\, dz.$$

Now apply (2.24). □

Lemma 2.7 (Tails at or above the inner interface)**.** *If $x_r \ge \mu = \rho K$ and $x_r < K$, then*

$$\mathcal{T}_r \le 2 \int_{x_r}^{K} A_{\rho,K}(z)\, dz.$$

Equality $x_r = \mu$ is included.

Proof. On $z \ge x_r \ge \mu$, one has $G_\mu(z) = 0$, including at $z = \mu$, so $A_{\rho,K}(z) = G_K(z)$. Set $b = K - x_r > 0$ and $g(t) = G_K(x_r + t)$ on $[0, b]$. Direct differentiation of (2.14) gives

$$G_K'(z) = -\frac{1}{\pi} \arccos \frac{z}{K} \in [-1/2, 0], \qquad G_K''(z) = \frac{1}{\pi\sqrt{K^2 - z^2}} > 0.$$

Thus g is nonnegative, decreasing, convex, and $1/2$-Lipschitz, with $g(b) = G_K(K) = 0$. Terms of (2.31) with $j > b$ vanish; if b is an integer, the terminal term is $\lfloor G_K(K) + 1/4 \rfloor = 0$. Therefore (2.30) is exactly the desired estimate. ☐

2.4 The low-interface estimate in the small-hole sector

Define

$$\omega_0 := G_1(1/2) = \frac{\sqrt{3}}{2\pi} - \frac{1}{6}, \qquad C_* := \frac{1}{2} + \frac{8\sqrt{2}}{15}. \tag{2.36}$$

Notice that $0 < \omega_0 < 1/2$. The lower inequality is equivalent to $\pi < 3\sqrt{3}$, which follows from $\pi < 4 < 3\sqrt{3}$; the upper inequality also follows directly from $0 < G_1(1/2) < G_1(0) = 1/\pi < 1/2$.

Lemma 2.8 (Low-interface shifted tails)**.** *Assume*

$$0 < \rho < \omega_0, \qquad K(\omega_0 - \rho) \ge C_*. \tag{2.37}$$

If $x_r = r + \frac{1}{2} < \mu = \rho K$, then, in fact strictly,

$$\mathcal{T}_r < 2 \int_{x_r}^{K} A_{\rho,K}(z)\, dz. \tag{2.38}$$

Proof. Fix a low tail and abbreviate $x_0 = x_r$, $A = A_{\rho,K}$. Put

$$n = \lfloor \mu - x_0 \rfloor, \qquad q = x_0 + n, \qquad B = \lceil K - x_0 \rceil. \tag{2.39}$$

Then

$$q \le \mu < q + 1, \qquad n < B, \qquad G_K(x_0 + B) = 0.$$

The point q is the largest half-integer in the tail grid that does not exceed μ; it need not equal $\lfloor \mu \rfloor$. A low tail also forces $\mu > x_0 \ge 1/2$, hence $\mu > 1/2$.

Let

$$h_A(t) = A(x_0 + t) + \frac{1}{4}, \qquad h_K(t) = G_K(x_0 + t) + \frac{1}{4}.$$

All samples at and beyond B have floor zero, so

$$\frac{\mathcal{T}_r}{2} = \mathbf{T}(h_A; 0, B).$$

If $n \ge 1$, additivity of the trapezoidal sum at the shared endpoint and $A \le G_K$ give

$$\frac{\mathcal{T}_r}{2} \le \mathbf{T}(h_A; 0, n) + \mathbf{T}(h_K; n, B). \tag{2.40}$$

At the split point, the two half-weights add to the original full sample; after replacing the suffix by G_K, its half-sample can only increase. If $n = 0$, there is no concave block, and instead

$$\frac{\mathcal{T}_r}{2} \le \mathbf{T}(h_K; 0, B). \tag{2.41}$$

For $0 < z < \mu$,

$$A''(z) = \frac{1}{\pi\sqrt{K^2 - z^2}} - \frac{1}{\pi\sqrt{\mu^2 - z^2}} < 0. \tag{2.42}$$

Thus A is concave on $[x_0, q]$, also when $q = \mu$ by continuity. For $n \ge 1$, the concave bound (2.27) gives

$$\mathbf{T}(h_A; 0, n) \le \int_0^n A(x_0 + t)\, dt + \frac{n}{4}. \tag{2.43}$$

For the suffix set $g(t) = G_K(x_0 + t)$, using the zero extension after K. It is nonnegative, decreasing, convex, globally $1/2$-Lipschitz, and $g(B) = 0$. Because $q \le \mu = \rho K < \omega_0 K < K/2$,

$$t_* := \frac{K}{2} - x_0$$

belongs to $[n, B]$ (strictly to the right of n). On $[t_*, B]$,

$$|g'(t)| \le \frac{1}{\pi} \arccos \frac{1}{2} = \frac{1}{3},$$

and

$$g(t_*) = G_K(K/2) = \omega_0 K.$$

Every hypothesis of (2.29) is therefore satisfied, and

$$\mathbf{T}(h_K; n, B) \le \int_n^B G_K(x_0 + t)\, dt - \frac{1}{4}\lfloor \omega_0 K \rfloor. \tag{2.44}$$

The identical statement with $n = 0$ applies to (2.41).

Combining (2.43) and (2.44), or using only the latter when $n = 0$, leaves one and only one integral mismatch:

$$\delta := \int_q^\mu G_\mu(z)\, dz. \tag{2.45}$$

Indeed,

$$\int_{x_0}^q A(z)\, dz + \int_q^K G_K(z)\, dz = \int_{x_0}^K A(z)\, dz + \delta,$$

and the same identity holds for $n = 0$, where $q = x_0$. Hence

$$\frac{\mathcal{T}_r}{2} \le \int_{x_0}^K A(z)\, dz + \delta - \frac{\lfloor \omega_0 K \rfloor - n}{4}. \tag{2.46}$$

The turning-point estimate of Lemma 6.2 in [3] states that

$$G_\mu(z) < \frac{(\mu - z)^{3/2}}{3\sqrt{\mu}}, \qquad 0 < z < \mu. \tag{2.47}$$

If $q < \mu$, integration gives a strict first bound; if $q = \mu$, both sides below vanish. The endpoint-safe statement is

$$0 \le \delta \le \frac{2(\mu - q)^{5/2}}{15\sqrt{\mu}} < \frac{2\sqrt{2}}{15}, \tag{2.48}$$

because $0 \le \mu - q < 1$ and $\mu > 1/2$. The last inequality remains strict when $q = \mu$, since then $\delta = 0$.

Finally, $x_0 \ge 1/2$ implies

$$n \le \mu - x_0 \le \rho K - \frac{1}{2},$$

while $\lfloor y \rfloor > y - 1$ for every real y. Therefore

$$\begin{aligned} \lfloor \omega_0 K \rfloor - n &> \omega_0 K - 1 - \left(\rho K - \frac{1}{2}\right) \\ &= K(\omega_0 - \rho) - \frac{1}{2} \\ &\ge \frac{8\sqrt{2}}{15} > 4\delta. \end{aligned}$$

Both strict inequalities remain strict when the threshold in (2.37) is attained. Substitution in (2.46) proves (2.38). □

2.5 Uniform closure of the small-hole endpoint

Set

$$\rho_* := \frac{\omega_0}{1 + 2C_*} = \frac{\frac{\sqrt{3}}{2\pi} - \frac{1}{6}}{2 + \frac{16\sqrt{2}}{15}}. \tag{2.49}$$

Since $\omega_0 > 0$ and $C_* > 0$, $0 < \rho_* < \omega_0$. Its defining identity is

$$\omega_0 - \rho_* = 2C_*\rho_*, \qquad \frac{\omega_0 - \rho_*}{2\rho_*} = C_*. \tag{2.50}$$

Theorem 2.9 (Complete small-hole endpoint)**.** *For every $0 < \rho \le \rho_*$ and every $K \ge 0$,*

$$\boxed{N_D(A_{\rho,1}, K^2) \le \frac{2}{9\pi}(1 - \rho^3)K^3.} \tag{2.51}$$

The face $\rho = \rho_*$, the interface wall $\rho K = 1/2$, and every strict spectral wall are included.

Proof. If $K = 0$, the strict spectrum is positive, so the left side and the right side of (2.51) are both zero. Assume $K > 0$ and write $\mu = \rho K$.

First suppose $\mu \le 1/2$. Every shifted-tail start satisfies

$$x_r = r + \frac{1}{2} \ge \frac{1}{2} \ge \mu.$$

Lemma 2.7 therefore proves (2.33) for every active tail. At the joining wall $\mu = 1/2$, the first start is exactly $x_0 = \mu$, which belongs to the inclusive high-interface lemma.

Now suppose $\mu > 1/2$. Then

$$K > \frac{1}{2\rho}. \tag{2.52}$$

For $0 < \rho \le \rho_*$, the defining identity (2.50) gives

$$\frac{\omega_0 - \rho}{2\rho} \ge C_*; \tag{2.53}$$

equivalently, multiply $(1+2C_*)\rho \le (1+2C_*)\rho_* = \omega_0$ by positive quantities. Since $\omega_0 - \rho > 0$, (2.52) and (2.53) imply

$$K(\omega_0 - \rho) > \frac{\omega_0 - \rho}{2\rho} \ge C_*. \tag{2.54}$$

Every start $x_r \geq \mu$ is controlled by Lemma 2.7; every start $x_r < \mu$ is controlled by Lemma 2.8, because $\rho \leq \rho_* < \omega_0$ and (2.54) supplies its threshold. Thus all active tails satisfy (2.33) in this branch as well.

Lemma 2.6 now yields

$$S_{\rho,K} \leq \frac{2}{9\pi}(1-\rho^3)K^3,$$

and Proposition 2.4 gives $N_D(A_{\rho,1}, K^2) \leq S_{\rho,K}$, proving (2.51).

At $\rho = \rho_*$, inequality (2.53) is an equality, but the first inequality in (2.54) is strict whenever $\mu > 1/2$. Hence the endpoint is included. If instead $K(\omega_0 - \rho) = C_*$, then

$$\mu = \frac{C_* \rho}{\omega_0 - \rho} \leq \frac{1}{2}$$

by (2.53), so the point already belongs to the all-high-interface branch. No limiting argument at $\rho = 0$, no continuity of Bessel roots, and no finite numerical verification is used. □

3 Product and complementary-action tools

We now prove the Pólya inequality uniformly as the shell becomes thin. In this section the counting function is strict:

$$N_D(\Omega, K^2) = \#\{j : \lambda_j(\Omega) < K^2\}.$$

We retain the global convention $[u]_< := \#\{n \in \mathbb{N} : n < u\} = \max\{0, \lceil u \rceil - 1\}$. Thus a spectral level lying exactly on the wall K^2 is not counted. Put

$$\varepsilon = 1 - \rho, \qquad a = \varepsilon K, \qquad m_\varepsilon = 1 - \varepsilon + \frac{\varepsilon^2}{3}. \tag{3.1}$$

Since $1 - \rho^3 = 3\varepsilon m_\varepsilon$, the desired right-hand side has the two equivalent forms

$$\frac{2}{9\pi}(1-\rho^3)K^3 = \frac{1}{\varepsilon^2}\frac{2a^3}{3\pi} m_\varepsilon. \tag{3.2}$$

The proof uses the optimized phase estimate already established in Proposition 2.4. We record precisely the consequence that is used here. For $\nu_\ell = \ell + \frac{1}{2}$, let $\gamma_{\rho,K}(\nu_\ell)$ be the normalized radial Bessel phase. If $0 \leq \nu_\ell \leq K$, that proposition gives

$$\gamma_{\rho,K}(\nu_\ell) < A_{\rho,K}(\nu_\ell) + \frac{1}{4}, \tag{3.3}$$

where, after the scaling in (3.1), $A_{\rho,K}(\nu) = \mathcal{A}(\varepsilon\nu)$ and $\mathcal{A}$ is the explicit action in (3.9) below. The exact radial count in channel ℓ is

$$\#\{n \geq 1 : n < \gamma_{\rho,K}(\nu_\ell)\}. \tag{3.4}$$

In particular, (3.3) is an inequality for the strict count; it is not a replacement of that count by an ordinary floor.

3.1 The product comparison

Under the standard unitary radial transformation, the angular channel $\ell \in \mathbb{N}_0$ is the Dirichlet operator

$$L_\ell = -\frac{d^2}{dr^2} + \frac{\ell(\ell+1)}{r^2} \quad \text{in } L^2((\rho, 1), dr),$$

and it occurs with multiplicity $2\ell + 1$. The following lemma includes the direction of the min–max comparison and every strict radial and angular wall.

Lemma 3.1 (Strict product majorant)**.** *Let $0 < \varepsilon < 1$, $a = \varepsilon K$, and define*

$$N(a) = \max\left\{0, \left\lceil \frac{a}{\pi} \right\rceil - 1\right\}. \tag{3.5}$$

For $1 \le n \le N(a)$, put

$$b_n = \sqrt{a^2 - n^2\pi^2}, \qquad M_n = \left\lceil \sqrt{\left(\frac{b_n}{\varepsilon}\right)^2 + \frac{1}{4}} - \frac{1}{2} \right\rceil.$$

Then

$$N_D(A_{\rho,1}, K^2) \le \mathcal{P}_\varepsilon(a) := \sum_{n=1}^{N(a)} M_n^2. \tag{3.6}$$

In particular, the strict shell count is zero for $0 \le a \le \pi$.

Proof. The two quadratic forms have the common domain $H_0^1(\rho, 1)$, and

$$\int_\rho^1 \left(|u'|^2 + \frac{\ell(\ell+1)}{r^2}|u|^2\right) dr \ge \int_\rho^1 \left(|u'|^2 + \ell(\ell+1)|u|^2\right) dr,$$

because $r \le 1$. The min–max principle therefore gives

$$\lambda_{\ell,n} \ge \left(\frac{n\pi}{\varepsilon}\right)^2 + \ell(\ell+1), \qquad n \ge 1.$$

Consequently, a shell eigenvalue in this channel can be counted below K^2 only if

$$n^2\pi^2 + \varepsilon^2\ell(\ell+1) < a^2. \tag{3.7}$$

The active radial integers are exactly those satisfying $n\pi < a$, whose number is (3.5). For a fixed active n, condition (3.7) is

$$\ell < \sqrt{\left(\frac{b_n}{\varepsilon}\right)^2 + \frac{1}{4}} - \frac{1}{2}.$$

Hence the allowed angular indices are $0, \dots, M_n - 1$, and their full spherical multiplicity is

$$\sum_{\ell=0}^{M_n-1} (2\ell+1) = M_n^2.$$

Summing proves (3.6).

If $a = m\pi$, the radial level $n = m$ is excluded by the strict inequality $n\pi < a$; thus $N(a) = m - 1$. At an angular wall $b_n^2/\varepsilon^2 = L(L+1)$, the displayed expression inside the ceiling is exactly L, so $M_n = L$ and the level $\ell = L$ is excluded. The jump of multiplicity $2L + 1$ occurs only above the wall. Finally, no positive integer satisfies $n\pi < a$ when $0 \le a \le \pi$, including both endpoints, so the product majorant and hence the shell count are zero there. $\square$

3.2 The complementary-action comparison

For $r \in \{\rho, 1\}$ and $0 \le y \le ar$, define

$$J_r(y) = \sqrt{a^2r^2 - y^2} - y \arccos \frac{y}{ar}, \tag{3.8}$$

and set

$$\mathcal{A}(y) = \begin{cases} \dfrac{J_1(y) - J_\rho(y)}{\pi\varepsilon}, & 0 \le y \le \rho a, \\ \dfrac{J_1(y)}{\pi\varepsilon}, & \rho a \le y \le a. \end{cases} \tag{3.9}$$

We extend $\mathcal{A}(y) = 0$ for $y > a$.

Lemma 3.2 (Action geometry and curvature switch)**.** *The function $\mathcal{A}$ is continuous, continuously differentiable at $y = \rho a$, and strictly decreasing from*

$$T := \mathcal{A}(0) = \frac{a}{\pi} \quad \textit{to} \quad \mathcal{A}(a) = 0.$$

Let $R : [0, T] \to [0, a]$ be its decreasing inverse, let

$$\tau = \mathcal{A}(\rho a), \qquad F(t) = R(t)^2, \qquad g(t) = -F'(t).$$

Then

$$\begin{gathered} g \textit{ decreases on } (0, \tau), \qquad g \textit{ increases on } (\tau, T), \\ F(0) = a^2, \qquad F(\tau) = \rho^2 a^2, \qquad F(T) = 0, \qquad g(T) = 2\pi\rho a, \end{gathered} \tag{3.10}$$

where $g(T)$ denotes the one-sided limit. Moreover $g(t) = O(t^{-1/3})$ as $t \downarrow 0$.

Proof. Direct differentiation on the appropriate open intervals gives

$$J_r'(y) = -\arccos \frac{y}{ar}, \qquad J_r''(y) = \frac{1}{\sqrt{a^2r^2 - y^2}}. \tag{3.11}$$

At $y = \rho a$, $J_\rho(\rho a) = 0$, and the two one-sided first derivatives of (3.9) both equal $-\arccos \rho/(\pi\varepsilon)$. Thus the two pieces and their first derivatives match. Both derivatives are negative away from $y = a$, so the inverse exists on exactly the stated interval.

On the outer branch $\rho a < y < a$, put $\alpha = \arccos(y/a)$. If $t = \mathcal{A}(y)$, then

$$g(t) = -\frac{2y}{\mathcal{A}'(y)} = \frac{2\pi\varepsilon y}{\alpha}. \tag{3.12}$$

The function y/α increases with y, whereas $y = R(t)$ decreases with t; hence g decreases on $0 < t < \tau$.

On the inner branch $0 < y < \rho a$, put

$$\delta(y) = \arccos \frac{y}{a} - \arccos \frac{y}{\rho a} > 0.$$

Differentiating $\arccos(y/(ar))$ with respect to r gives the exact identity

$$\frac{\delta(y)}{y} = \int_\rho^1 \frac{dr}{r\sqrt{a^2r^2 - y^2}}. \tag{3.13}$$

The integrand increases strictly with y, so $\delta(y)/y$ increases, and $y/\delta(y)$ decreases, with y. Since

$$g(t) = \frac{2\pi\varepsilon y}{\delta(y)}$$

and $y = R(t)$ decreases, g increases on $\tau < t < T$. Formula (3.12) and its inner counterpart agree at the interface.

The three values of F follow from $R(0) = a$, $R(\tau) = \rho a$, and $R(T) = 0$. As $y \downarrow 0$, (3.13) gives

$$\delta(y) = \frac{\varepsilon}{\rho a} y + O(y^3),$$

and therefore $g(T) = 2\pi\rho a$.

Finally, on the outer branch write $y = a\cos\theta$. Then

$$t = \frac{a}{\pi\varepsilon}(\sin\theta - \theta\cos\theta) \sim \frac{a\theta^3}{3\pi\varepsilon}, \qquad g(t) = \frac{2\pi\varepsilon a\cos\theta}{\theta} = O(t^{-1/3}). \tag{3.14}$$

This proves the asserted improper endpoint behavior. All square roots and arccosines in the proof are taken on their stated closed domains; every division occurs in an open interval where its denominator is positive, and the endpoint zeros are handled by the limits just computed. □

Lemma 3.3 (Strict phase-to-action layer cake)**.** *For $\ell \in \mathbb{N}_0$, set*

$$y_\ell = \varepsilon\left(\ell + \frac{1}{2}\right), \qquad q_\ell = \left[\mathcal{A}(y_\ell) + \frac{1}{4}\right]_<, \qquad P_{\mathcal{A}} = \varepsilon \sum_{\ell \geq 0} y_\ell q_\ell. \tag{3.15}$$

Then

$$N_D(A_{\rho,1}, K^2) \leq \frac{2P_{\mathcal{A}}}{\varepsilon^2}. \tag{3.16}$$

For $n \geq 1$, put $x_n = n - \frac{1}{4}$, and define

$$M_\varepsilon(x) = \#\left\{\ell \in \mathbb{N}_0 : \varepsilon\left(\ell + \frac{1}{2}\right) < x\right\} = \max\left\{0, \left\lceil \frac{x}{\varepsilon} - \frac{1}{2} \right\rceil\right\}. \tag{3.17}$$

The proxy has the exact finite representation

$$P_{\mathcal{A}} = \frac{\varepsilon^2}{2} \sum_{\substack{n \geq 1 \\ x_n < T}} M_\varepsilon(R(x_n))^2. \tag{3.18}$$

Moreover,

$$I := \frac{1}{2}\int_0^T F(t)\,dt = \frac{a^3}{3\pi}\left(1 - \varepsilon + \frac{\varepsilon^2}{3}\right), \qquad \frac{2I}{\varepsilon^2} = \frac{2}{9\pi}(1-\rho^3)K^3. \tag{3.19}$$

Proof. For $y_\ell < a$, equations (3.3) and (3.4) show that the number of radial levels in channel ℓ is at most q_ℓ. If $y_\ell \geq a$, equivalently $\nu_\ell \geq K$, the product comparison gives

$$\lambda_{\ell,1} \geq \frac{\pi^2}{\varepsilon^2} + \ell(\ell+1) = \frac{\pi^2}{\varepsilon^2} + \nu_\ell^2 - \frac{1}{4} > K^2,$$

so there is no radial level; at $y_\ell = a$, the definition also gives $q_\ell = [1/4]_< = 0$. Multiplicity $2\ell + 1 = 2y_\ell/\varepsilon$ now yields

$$N_D(A_{\rho,1}, K^2) \leq \sum_{\ell \geq 0} \frac{2y_\ell}{\varepsilon} q_\ell = \frac{2P_{\mathcal{A}}}{\varepsilon^2},$$

which proves (3.16). If $\mathcal{A}(y_\ell) + 1/4 = m \in \mathbb{Z}$, then $q_\ell = m - 1$, so the phase-proxy wall is excluded exactly as required.

For every $y \geq 0$, the strict bracket is

$$\left[\mathcal{A}(y) + \frac{1}{4}\right]_< = \#\{n \geq 1 : x_n < \mathcal{A}(y)\}.$$

The sums are finite. Since $\mathcal{A}$ and R are decreasing, $x_n < \mathcal{A}(y_\ell)$ is equivalent to $y_\ell < R(x_n)$. Therefore

$$\begin{aligned} P_{\mathcal{A}} &= \varepsilon \sum_{\substack{n\geq 1\\ x_n<T}} \sum_{y_\ell < R(x_n)} y_\ell \\ &= \varepsilon \sum_{\substack{n\geq 1\\ x_n<T}} \varepsilon \sum_{\ell=0}^{M_\varepsilon(R(x_n))-1} \left(\ell + \frac{1}{2}\right) \\ &= \frac{\varepsilon^2}{2} \sum_{\substack{n\geq 1\\ x_n<T}} M_\varepsilon(R(x_n))^2, \end{aligned}$$

because $\sum_{\ell=0}^{M-1}(\ell + 1/2) = M^2/2$. This proves (3.18). Formula (3.17) follows from the same strict counting; at $x/\varepsilon = m + 1/2$, it equals m, excluding $\ell = m$.

Differentiation of (3.8) with respect to r gives

$$\frac{\partial J_r(y)}{\partial r} = \sqrt{a^2 - \frac{y^2}{r^2}} \quad (0 \leq y \leq ar).$$

Consequently the two cases of (3.9) combine as

$$\mathcal{A}(y) = \frac{1}{\pi\varepsilon} \int_\rho^1 \left(a^2 - \frac{y^2}{r^2}\right)_+^{1/2} dr, \tag{3.20}$$

where $s_+ = \max\{s, 0\}$. The area identity for a decreasing inverse and Tonelli's theorem give

$$\frac{1}{2}\int_0^T R(t)^2\,dt = \int_0^a y\mathcal{A}(y)\,dy.$$

Using (3.20) and then $y = rz$,

$$\begin{aligned} \int_0^a y\mathcal{A}(y)\,dy &= \frac{1}{\pi\varepsilon}\int_\rho^1 r^2\,dr \int_0^a z\sqrt{a^2 - z^2}\,dz \\ &= \frac{a^3}{3\pi\varepsilon}\int_\rho^1 r^2\,dr = \frac{a^3}{3\pi}\left(1 - \varepsilon + \frac{\varepsilon^2}{3}\right). \end{aligned}$$

This is (3.19); its final equality follows from $1 - \rho^3 = 3\varepsilon m_\varepsilon$ and $K = a/\varepsilon$. $\square$

Lemma 3.4 (Shifted sawtooth and radial quadrature deficit)**.** *Let*

$$C(t) = \#\{n \geq 1 : n - \tfrac{1}{4} < t\}, \qquad w(t) = t - C(t), \qquad \mathfrak{W}(t) = \int_0^t w(s)\,ds,$$

and put $\Psi(t) = \mathfrak{W}(t) - t/4$. *Then, for all* $t \geq 0$,

$$\mathfrak{W}(t) \geq 0, \qquad -\frac{1}{32} \leq \Psi(t) \leq \frac{3}{32}. \tag{3.21}$$

Furthermore, with

$$D_{\mathrm{rad}} = \int_0^T F(t)\,dt - \sum_{\substack{n\geq 1\\ x_n<T}} F(x_n), \tag{3.22}$$

one has

$$D_{\mathrm{rad}} \geq \frac{\rho^2 a^2}{4} - \frac{\pi\rho a}{4}. \tag{3.23}$$

Proof. On $0 \le t \le 3/4$, $C(t) = 0$, and hence

$$\Psi(t) = \frac{1}{2}\left(t - \frac{1}{4}\right)^2 - \frac{1}{32}. \tag{3.24}$$

For $x_n = n - 1/4$, $n \ge 1$, and $t = x_n + u$ with $0 \le u \le 1$, direct integration gives

$$\Psi(t) = \frac{1}{2}\left(u - \frac{1}{2}\right)^2 - \frac{1}{32}. \tag{3.25}$$

These formulas agree at cell endpoints and prove the two-sided bound for Ψ. On the first cell $\mathfrak{W} = t^2/2 \ge 0$. Thereafter $\mathfrak{W} = t/4 + \Psi \ge t/4 - 1/32 > 0$, proving (3.21). At a jump x_n, strict counting gives the one-sided values $w(x_n^-) = 3/4$ and $w(x_n^+) = -1/4$, while $\mathfrak{W}$ and Ψ are continuous.

Distributionally, $dw = dt - dC$. Since $F(T) = 0$, an atom at $x_n = T$ has zero weight and is consistent with its exclusion from (3.22). Stieltjes integration by parts, first on $[\eta, T]$ and then with $\eta \downarrow 0$, gives

$$D_{\mathrm{rad}} = \int_0^T w(t)g(t)\,dt. \tag{3.26}$$

Indeed, (3.14), together with $w(t) = t$ and $\mathfrak{W}(t) = t^2/2$ near zero, gives $w(t)g(t) = O(t^{2/3})$ and $\mathfrak{W}(t)g(t) = O(t^{5/3}) \to 0$, so the improper boundary term vanishes.

Split (3.26) at the actual value τ, with no assumption that τ belongs to any particular sawtooth cell. On $(0, \tau)$, g decreases and $\mathfrak{W} \ge 0$; hence

$$\int_0^\tau wg = \mathfrak{W}(\tau)g(\tau) - \int_0^\tau \mathfrak{W}\,dg \ge \mathfrak{W}(\tau)g(\tau). \tag{3.27}$$

On (τ, T), g increases and $w = 1/4 + \Psi'$ almost everywhere. Adding its integration-by-parts identity to (3.27), and using $\mathfrak{W}(\tau) = \tau/4 + \Psi(\tau)$, cancels the interface value and yields

$$D_{\mathrm{rad}} \ge \frac{F(\tau)}{4} + \frac{\tau g(\tau)}{4} + \Psi(T)g(T) - \int_\tau^T \Psi\,dg.$$

Because dg is a positive measure on (τ, T), the bounds (3.21) imply

$$\begin{aligned} D_{\mathrm{rad}} &\ge \frac{F(\tau)}{4} - \frac{g(T)}{8} + g(\tau)\left(\frac{\tau}{4} + \frac{3}{32}\right) \\ &\ge \frac{F(\tau)}{4} - \frac{g(T)}{8}. \end{aligned}$$

The two discarded quantities, $-\int_0^\tau \mathfrak{W}\,dg$ and $g(\tau)(\tau/4 + 3/32)$, are explicitly nonnegative. Substitution of $F(\tau) = \rho^2 a^2$ and $g(T) = 2\pi\rho a$ from Lemma 3.2 proves (3.23). Continuity of $\mathfrak{W}, \Psi, g$ at the split shows that the same proof covers τ below, on, or above every sawtooth grid point. □

4 The compact-middle argument in full detail

We retain the strict counting convention $[x]_< := \#\{n \in \mathbb{N} : n < x\}$, put

$$W(\rho, K) := \frac{2}{9\pi}(1 - \rho^3)K^3, \qquad z_\rho := \frac{\pi}{1 - \rho}, \qquad k_m(\rho) := \sqrt{z_\rho^2 + m(m+1)},$$

and denote by $q_{\ell,n}(\rho)$ the n-th positive frequency in angular channel ℓ. Its multiplicity is $2\ell + 1$. The phase reduction proved earlier gives

$$N_D(A_{\rho,1}, K^2) \le P_{\mathrm{coarse}}(\rho, K) := \sum_{\ell + 1/2 < K} (2\ell + 1)\left[A_{\rho,K}(\ell + 1/2) + \frac{1}{4}\right]_<, \tag{4.1}$$

where $A_{\rho,K} = G_K - G_{\rho K}$ and

$$G_a(\nu) = \begin{cases} \dfrac{1}{\pi}\left(\sqrt{a^2-\nu^2} - \nu \arccos \dfrac{\nu}{a}\right), & 0 \le \nu < a, \\ 0, & \nu \ge a. \end{cases}$$

The value at both interfaces is therefore fixed by zero extension. This will matter at $\nu = \rho K$, at $\nu = K$, and at every integer proxy wall.

4.1 Constants and the primitive analytic envelope

For $0 \le s \le 1$, set

$$G_1(s) = \frac{1}{\pi}\left(\sqrt{1-s^2} - s \arccos s\right), \quad \omega_0 = G_1(1/2) = \frac{\sqrt{3}}{2\pi} - \frac{1}{6},$$

$$C_* = \frac{1}{2} + \frac{8\sqrt{2}}{15}, \qquad C_0 = 1 + \frac{8\sqrt{2}}{15}, \qquad \rho_* := \frac{\omega_0}{1+2C_*} = \frac{\omega_0}{2C_0}.$$

For $0 < \rho < 1$, define

$$a_\rho = \frac{2\pi\rho}{1-\rho}, \qquad \eta_\rho = G_1(\max\{\rho, 1/2\}),$$

$$K_0(\rho) = \left(\frac{\sqrt{a_\rho} + \sqrt{a_\rho + 4\eta_\rho C_0}}{2\eta_\rho}\right)^2, \qquad H_0(\rho) = \frac{C_*}{\omega_0 - \rho} \quad (\rho < \omega_0). \tag{4.2}$$

We first re-establish every primitive owner that is needed to form the compact residual. This prevents the residual description from hiding an unproved coverage assumption.

Proposition 4.1 (Primitive analytic owners)**.** *The Pólya bound holds on each of the following inclusive domains:*

1. $2\rho K \le 1$*;*
2. $(1-\rho)K \le \pi$*;*
3. $0 < \rho < \omega_0$ *and* $K(\omega_0 - \rho) \ge C_*$*, equivalently* $K \ge H_0(\rho)$*;*
4. $0 < \rho \le \rho_*$*, for every* $K \ge 0$*;*
5. $K \ge K_0(\rho)$*;*

Proof. For item 1, $\mu = \rho K \le 1/2$; every shifted half-integer tail begins at or above the inner interface and is controlled by the convex high-angular tail inequality. Summing the tail integrals with the weighted identity gives $N_D \le W$. For item 2, $K \le z_\rho = q_{0,1}$, and (4.7) (whose elementary proof is repeated below) shows that no radial mode is strictly below K.

We record the tail bookkeeping used in items 3–5. Put

$$x_r = r + \frac{1}{2}, \qquad q_\ell = \left\lfloor A_{\rho,K}\left(\ell + \frac{1}{2}\right) + \frac{1}{4}\right\rfloor$$

and

$$\mathcal{T}_r = \left\lfloor A_{\rho,K}(x_r) + \frac{1}{4}\right\rfloor + 2\sum_{j\ge 1}\left\lfloor A_{\rho,K}(x_r + j) + \frac{1}{4}\right\rfloor.$$

All sums are finite because $A_{\rho,K} = 0$ at and beyond K. For integers $a < b$, write

$$\mathbf{T}(h,a,b) = \frac{1}{2}\lfloor h(a)\rfloor + \sum_{j=a+1}^{b-1}\lfloor h(j)\rfloor + \frac{1}{2}\lfloor h(b)\rfloor.$$

The elementary multiplicity identity

$$\sum_{\ell\ge 0}(2\ell+1)q_\ell=\sum_{r\ge 0}\left(q_r+2\sum_{\ell>r}q_\ell\right)=\sum_{r\ge 0}\mathcal{T}_r$$

shows that $P_{\mathrm{coarse}}\le\sum_r\mathcal{T}_r$. If every active tail satisfies

$$\mathcal{T}_r\le 2\int_{x_r}^{K}A_{\rho,K}(z)\,dz,$$

then, with $f_3(z)=\#\{r\ge 0: x_r\le z\}$ and $F_3(z)=\int_0^z f_3(t)\,dt$,

$$P_{\mathrm{coarse}}\le 2\int_0^K f_3(z)A_{\rho,K}(z)\,dz.$$

Writing $z=m+1/2+s$, $m\in\mathbb{N}_0$, $0\le s<1$, gives

$$\frac{z^2}{2}-F_3(z)=\frac{1}{2}\left(s-\frac{1}{2}\right)^2\ge 0;$$

for $0\le z<1/2$, the same inequality is immediate. Since $A_{\rho,K}$ is decreasing, absolutely continuous, and vanishes at K, integration by parts yields

$$P_{\mathrm{coarse}}\le -\int_0^K z^2A'_{\rho,K}(z)\,dz=\int_0^K 2zA_{\rho,K}(z)\,dz=\frac{2}{9\pi}(1-\rho^3)K^3=W(\rho,K).$$

For the final equality, scaling reduces each G_a-integral to

$$\int_0^1 2xG_1(x)\,dx=\frac{2}{\pi}\left(\frac{1}{3}-\frac{2}{9}\right)=\frac{2}{9\pi}.$$

We shall use the exact trapezoidal-floor estimates proved in the one-dimensional reduction. For a nonnegative concave h, $\mathbf{T}(h,a,b)\le\int_a^b h(t)\,dt$. If g is nonnegative, decreasing, convex, and 1/2-Lipschitz, vanishes at b, and is 1/3-Lipschitz after t_*, then

$$\mathbf{T}(g+1/4,a,b)\le\int_a^b g(t)\,dt-\frac{1}{4}\lfloor g(t_*)\rfloor.$$

The corresponding shifted convex estimate gives $\mathcal{T}_r\le 2\int_{x_r}^K G_K(z)\,dz$ whenever $x_r\ge\mu$. Indeed, on that interval $A=G_K$, and

$$G'_K(z)=-\frac{1}{\pi}\arccos\frac{z}{K}\in[-1/2,0],\qquad G''_K(z)=\frac{1}{\pi\sqrt{K^2-z^2}}>0.$$

We now prove item 3. Write $A=A_{\rho,K}$. Only a start $x_0<\mu=\rho K$ remains. Put

$$n=\lfloor\mu-x_0\rfloor,\qquad q=x_0+n,\qquad B=\lceil K-x_0\rceil.$$

Thus $q\le\mu<q+1$, $n<B$, and zero extension gives $G_K(x_0+B)=0$. The action is concave on $[x_0,q]$, because

$$A''(z)=\frac{1}{\pi\sqrt{K^2-z^2}}-\frac{1}{\pi\sqrt{\mu^2-z^2}}<0\qquad(0<z<\mu).$$

Moreover $q\le\mu=\rho K<\omega_0K<K/2$. Splitting at n, using the concave estimate on the head and the improved convex estimate on the G_K-suffix with $t_*=K/2-x_0$, gives, also when $n=0$,

$$\frac{\mathcal{T}_r}{2}\le\int_{x_0}^q A(z)\,dz+\frac{n}{4}+\int_q^K G_K(z)\,dz-\frac{1}{4}\lfloor\omega_0K\rfloor.$$

The only integral mismatch is

$$\delta := \int_q^\mu G_\mu(z)\,dz.$$

To bound it, write $t = \cos\theta$. Concavity of sin on $[0, \pi/4]$ gives $\sin(\theta/2) \geq \sqrt{2}\,\theta/\pi$, hence $\arccos t \leq (\pi/2)\sqrt{1-t}$ for $0 \leq t \leq 1$. Since

$$G_\mu(z) = \frac{1}{\pi}\int_z^\mu \arccos\frac{t}{\mu}\,dt,$$

we obtain

$$0 \leq \delta \leq \frac{2(\mu-q)^{5/2}}{15\sqrt{\mu}} < \frac{2\sqrt{2}}{15},$$

where a low tail has $\mu > x_0 \geq 1/2$ and $0 \leq \mu - q < 1$. Consequently

$$\frac{\mathcal{T}_r}{2} \leq \int_{x_0}^K A(z)\,dz + \delta - \frac{\lfloor \omega_0 K\rfloor - n}{4}.$$

Finally $n \leq \rho K - 1/2$ and $\lfloor y\rfloor > y - 1$, so

$$\lfloor \omega_0 K\rfloor - n > K(\omega_0 - \rho) - \frac{1}{2} \geq C_* - \frac{1}{2} = \frac{8\sqrt{2}}{15} > 4\delta.$$

Thus every low-interface tail is strict, every high-interface tail is non-strict, and the weighted scaffold proves item 3, including the face $K(\omega_0 - \rho) = C_*$.

For item 4, if $\mu \leq 1/2$ use item 1. Otherwise $K > 1/(2\rho)$. Because $\rho_* = \omega_0/(1 + 2C_*)$, for $0 < \rho \leq \rho_*$,

$$K(\omega_0 - \rho) > \frac{\omega_0 - \rho}{2\rho} \geq \frac{\omega_0 - \rho_*}{2\rho_*} = C_*,$$

and item 3 applies. All faces are included; at $K = 0$ both sides vanish.

For item 5, fix a low-interface tail $x_0 = r + 1/2 < \mu$, retain the preceding definition of $\mathcal{T}_r$, and write $A = A_{\rho,K}$. Put $n = \lfloor \mu - x_0\rfloor$, $q = x_0 + n$, and $B = \lceil K - x_0\rceil$. On $[0, n]$, $h(t) = A(x_0 + t) + 1/4$ is decreasing, concave, and Lipschitz with constant $c_\rho = \arccos\rho/\pi$. If p is the last index of its first constant ordinary-floor shelf (put $p = n$ when the shelf never drops), the concave trapezoidal-floor inequality gives

$$\mathbf{T}(h, 0, n) \leq \int_0^n A(x_0 + t)\,dt + \frac{n}{4} - \frac{1 - c_\rho}{2}(n - p). \tag{4.3}$$

For $n = 0$, the head is absent and $p = 0$.

Let $\vartheta = -A'$. Direct differentiation gives

$$\vartheta'(z) = \frac{1}{\pi\sqrt{\mu^2 - z^2}} - \frac{1}{\pi\sqrt{K^2 - z^2}} \geq \frac{1 - \rho}{\pi\rho K} =: \kappa_\rho.$$

Equal floors on the first shelf imply $0 \leq A(x_0) - A(x_0 + p) < 1$, including at an integer endpoint. Hence

$$1 > \int_{x_0}^{x_0+p} \vartheta(z)\,dz \geq \frac{\kappa_\rho p^2}{2}, \qquad p < \sqrt{a_\rho K}.$$

For the convex tail set $s = \max\{q, K/2\}$. The function $G_K(x_0 + t)$ is decreasing and convex, is Lipschitz $1/2$ on the whole tail and $1/3$ after s, and $G_K(s) \geq K\eta_\rho$. The convex trapezoidal-floor inequality therefore gives

$$\mathbf{T}(G_K(x_0 + \cdot) + 1/4, n, B) \leq \int_n^B G_K(x_0 + t)\,dt - \frac{1}{4}\lfloor K\eta_\rho\rfloor. \tag{4.4}$$

At $j = n$, the two half endpoint weights in (4.3)–(4.4) majorize the original full A-sample. Their only integral mismatch is

$$\delta = \int_q^\mu G_\mu(z)\,dz, \qquad 0 \le \delta \le \frac{2(\mu-q)^{5/2}}{15\sqrt{\mu}} < \frac{2\sqrt{2}}{15}.$$

Writing $d_\rho = 1 - 2c_\rho = 2\arcsin\rho/\pi$, combination of the head and tail yields

$$\frac{\mathcal{T}_r}{2} \le \int_{x_0}^K A(z)\,dz + \delta - \frac{M}{4}, \quad M = \lfloor K\eta_\rho \rfloor + d_\rho(n-p) - p.$$

Since $d_\rho > 0$ and $p < \sqrt{a_\rho K}$, the strict floor inequality, valid even when $K\eta_\rho$ is an integer, gives

$$M > K\eta_\rho - 1 - \sqrt{a_\rho K}.$$

Thus the exact sufficient condition is

$$K\eta_\rho - \sqrt{a_\rho K} \ge C_0. \tag{4.5}$$

Indeed it implies $M > 8\sqrt{2}/15 > 4\delta$, so every low-interface tail is strictly below twice its action integral. Tails beginning at or above μ satisfy the non-strict convex high-interface bound; the weighted identity completes the proof. The cases $n = 0$, $p = 0$, $p = n$, $q = \mu$, q on either side of $K/2$, and all strict-floor walls are included in the displayed inequalities.

Writing $Y = \sqrt{K}$, the left side minus C_0 is $\eta_\rho Y^2 - \sqrt{a_\rho}Y - C_0$. It has one positive root, and the square of that root is precisely $K_0(\rho)$ in (4.2). Hence (4.5) holds, including equality, exactly for $K \ge K_0(\rho)$.

□

Proposition 4.1 supplies the inclusive frequency owners needed in the open ratio strip $\rho_* < \rho < 39/50$:

$$2\rho K \le 1, \qquad (1-\rho)K \le \pi, \qquad \rho < \omega_0,\ K \ge H_0(\rho), \qquad K \ge K_0(\rho).$$

The endpoint $\rho = \rho_*$ is already owned by Theorem 2.9; the opposite endpoint $\rho = 39/50$ will be owned by the all-frequency optical theorem.

Put

$$L(\rho) = \max\left\{\frac{1}{2\rho}, \frac{\pi}{1-\rho}\right\}, \qquad U(\rho) = \min\big(\{K_0(\rho)\} \cup \{H_0(\rho) : \rho < \omega_0\}\big).$$

Since every primitive frequency wall is included, exact complementation within this ratio strip gives

$$\mathcal{D}_{16} = \left\{(\rho, K) : \rho_* < \rho < \frac{39}{50}, \quad L(\rho) < K < U(\rho)\right\}. \tag{4.6}$$

Both frequency inequalities are strict. No numerical locator for the switch between H_0 and K_0 is needed; the minimum definition retains the common face automatically.

The two lower walls meet at

$$\rho_c := \frac{1}{1+2\pi}, \qquad L(\rho) = \begin{cases} (2\rho)^{-1}, & \rho \le \rho_c, \\ z_\rho, & \rho \ge \rho_c. \end{cases}$$

4.2 Variational comparisons and the complete zero registry

The transformed channel form is

$$\mathfrak{a}_\ell[u] = \int_\rho^1 \left(|u'|^2 + \frac{\ell(\ell+1)}{r^2}|u|^2 \right) dr, \qquad D(\mathfrak{a}_\ell) = H_0^1(\rho, 1).$$

Since $r^{-2} \geq 1$, the min–max principle gives, at every radial index,

$$q_{\ell,n}^2 \geq n^2 z_\rho^2 + \ell(\ell+1), \qquad q_{0,n} = n z_\rho. \tag{4.7}$$

Extension by zero from $(\rho, 1)$ to $(0, 1)$ embeds the shell fixed-channel form domain isometrically into the ball fixed-channel form domain. The zero trace at $r = \rho$ preserves membership in $H_0^1(0, 1)$, and Hardy's inequality controls the centrifugal term. Thus, preserving both ℓ and n,

$$q_{\ell,n}^2 \geq j_{\ell+1/2,n}^2. \tag{4.8}$$

The ball forms have the same domain for every angular order; hence, for $p \geq \ell$,

$$j_{p+1/2,n}^2 \geq j_{\ell+1/2,n}^2 + p(p+1) - \ell(\ell+1). \tag{4.9}$$

These are fixed-channel comparisons, not an invocation of unlabelled global domain monotonicity.

We use the exact elementary enclosure

$$\frac{333}{106} < \pi < \frac{22}{7}. \tag{4.10}$$

For the lower bound, Machin's identity $\pi = 16\arctan(1/5) - 4\arctan(1/239)$ and alternating sums give a strictly stronger rational bound; for the upper bound,

$$0 < \int_0^1 \frac{x^4(1-x)^4}{1+x^2}\, dx = \frac{22}{7} - \pi.$$

For clarity we now derive every specialized Bessel-zero inequality used below, apart from the one published first-zero estimate that is stated explicitly before specialization. The spherical Bessel identity and recurrences are standard [5]. Set

$$\mathrm{j}_\ell(x) = \sqrt{\frac{\pi}{2x}} J_{\ell+1/2}(x), \qquad P_\ell(x) = x^{\ell+1}\mathrm{j}_\ell(x).$$

Then

$$P_{\ell+1} = (2\ell+1)P_\ell - x^2 P_{\ell-1}, \qquad P_\ell' = x P_{\ell-1}, \tag{4.11}$$

with $P_0 = \sin x$ and $P_1 = \sin x - x\cos x$. In particular,

$$\begin{aligned}
P_2 &= (3 - x^2)\sin x - 3x\cos x, \\
P_3 &= (15 - 6x^2)\sin x + (x^3 - 15x)\cos x, \\
P_5 &= (945 - 420x^2 + 15x^4)\sin x + (-945x + 105x^3 - x^5)\cos x, \\
P_6 &= (10395 - 4725x^2 + 210x^4 - x^6)\sin x \\
&\quad + (-10395x + 1260x^3 - 21x^5)\cos x, \\
P_7 &= (135135 - 62370x^2 + 3150x^4 - 28x^6)\sin x \\
&\quad + (-135135x + 17325x^3 - 378x^5 + x^7)\cos x.
\end{aligned}$$

Positive zeros are simple by ODE uniqueness. At a zero of j_ℓ, the derivative recurrence gives $\mathrm{j}_{\ell+1} = -\mathrm{j}_\ell'$. Combined with (4.9), this gives strict interlacing and makes the following sign/count ledger a complete enumeration, rather than a sampled sign test:

endpoint	half-period	signs $\mathrm{j}_0, \ldots, \mathrm{j}_\ell$	zero counts $N_0, \ldots, N_\ell$
77/10	$(2\pi, 5\pi/2)$	$+, -$	$2, 1$
9	$(5\pi/2, 3\pi)$	$+, +, -$	$2, 2, 1$
10	$(3\pi, 7\pi/2)$	$-, +, +, -, -, -, +$	$3, 2, 2, 1, 1, 1, 0$
103/10	$(3\pi, 7\pi/2)$	$-, +, +, -$	$3, 2, 2, 1$
21/2	$(3\pi, 7\pi/2)$	$-, +$	$3, 2$
23/2	$(7\pi/2, 4\pi)$	$-, -, +, +, -, -, -, +$	$3, 3, 2, 2, 1, 1, 1, 0$
61/5	$(7\pi/2, 4\pi)$	$-, -, +$	$3, 3, 2$
129/10	$(4\pi, 9\pi/2)$	$+, -, -, +, +, -$	$4, 3, 3, 2, 2, 1$

For example, the endpoint signs follow from (4.10), the displayed polynomials, and $\tan s > s$, $\tan s > s + s^3/3$, $\tan s < s/(1 - s^2/2)$ for $0 < s < 1$. The nonautomatic strict comparisons can be taken as

77/10	$5\pi/2 - x > 3/20 > 10/77$
21/2	$7\pi/2 - x > 49/100 > 2/21$
9	$3\pi - x > 21/50 > 9/26$
61/5	$4\pi - x > 9/25 > 915/3646$
103/10	$7\pi/2 - x > 69/100 > 621540/938227$
129/10	$0 < x - 4\pi < 17/50, \quad \tan(x - 4\pi) < 1700/4711$
10	$4/7 < x - 3\pi < 29/50, \quad \tan(x - 3\pi) < 2900/4159$
23/2	$53/50 < 4\pi - x < \pi/2, \quad \tan(4\pi - x) > 546377/375000.$

All comparisons are between positive rationals and are checked by cross multiplication. The ledger proves

$$\begin{gathered} j_{3/2,2} > 77/10, \quad j_{5/2,2} > 9, \quad j_{7/2,2} > 103/10, \quad j_{11/2,2} > 129/10, \\ j_{3/2,3} > 21/2, \quad j_{5/2,3} > 61/5, \\ j_{13/2,1} > 10, \quad j_{15/2,1} > 23/2. \end{gathered} \tag{4.12}$$

This includes the requested higher-radial derivations. For additional detail, the second root of P_3 is the unique root in $(3\pi, 7\pi/2)$: the derivative chain $R' = x \sin x$, $P_2' = xR$, $P_3' = xP_2$ first excludes every earlier cell, and at $x = 103/10$, writing $y = x - 3\pi$, alternating Taylor bounds give $\tan y < 5/4$, while

$$\frac{x(x^2 - 15)}{6x^2 - 15} - \frac{3}{2} = \frac{5917}{621540} > 0,$$

so $P_3(x) < 0$. Likewise, the third root of P_2 is unique in $(7\pi/2, 4\pi)$; at $x = 61/5$,

$$4\pi - x > \frac{97}{265} > \frac{3x}{x^2 - 3},$$

so $P_2(x) > 0$ before its decreasing zero. At $x = 129/10$, the explicit P_5 coefficients and $\tan(x - 4\pi) < 3/8 < |B|/A$ give $P_5(x) < 0$; angular propagation of $j_{5/2,3} > 61/5$ puts the third P_5-zero above this point, hence its second zero is also above it. These arguments use no root table.

For the remaining first zeros we use Lorch's strict first-zero inequality [4]

$$j_{\nu,1}^2 > \frac{24(\nu+1)^2}{1 - 2\nu + \sqrt{(2\nu+3)(2\nu+11)}} - 2(\nu^2 - 1), \qquad \nu > -1.$$

At $\nu = \ell + 1/2$, its right side is

$$R_\ell = \frac{3(2\ell+3)\sqrt{(\ell+2)(\ell+6)} - 2\ell^2 + \ell + 6}{4}.$$

Exact positive-side squaring gives

ℓ	first-zero wall	integer residual proving $R_\ell > c^2$
2	51/10	$105/4 - (51/10)^2 = 6/25$
3	13/2	$81^2 \cdot 5 - 178^2 = 1121$
4	15/2	$66^2 \cdot 15 - 247^2 = 4331$
5	87/10	198189
8	71/6	$1026^2 \cdot 35 - 6067^2 = 35171$
9	64/5	$1575^2 \cdot 165 - 20059^2 = 6939644$
10	69/5	$3450^2 \cdot 3 - 5911^2 = 767579.$

Together with (4.12), this yields the complete first-mode walls

$$\begin{array}{c|cccccccccc} \ell & 1 & 2 & 3 & 4 & 5 & 6 & 7 & 8 & 9 & 10 \\ \hline \beta_{\ell,1} & 4 & 51/10 & 13/2 & 15/2 & 87/10 & 10 & 23/2 & 71/6 & 64/5 & 69/5. \end{array} \tag{4.13}$$

The $\ell = 1$ wall follows internally from $\tan x = x$: the first positive root lies in $(\pi, 3\pi/2)$, and $\tan 4 < 2 < 4$. Applying (4.9) to the higher-radial seeds gives, for later use,

$$j^2_{p+1/2,2} > 9409/100 + p(p+1) \qquad (p \ge 3), \tag{4.14}$$

$$j^2_{p+1/2,3} > 3571/25 + p(p+1) \qquad (p \ge 2), \tag{4.15}$$

$$j^2_{p+1/2,2} > 13641/100 + p(p+1) \qquad (p \ge 5). \tag{4.16}$$

Equations (4.8)–(4.16) transfer every displayed wall to the shell at the same radial index.

4.3 The two-sided staircase

Put

$$\rho_0 = \frac{1}{\sqrt{337}}, \qquad d = \frac{\sqrt{337}}{2}, \qquad p_1(\rho) = \sqrt{4z_\rho^2 + 2}.$$

Lemma 4.2 (High side through k_6)**.** *For $\rho_c \le \rho \le 39/50$ and $z_\rho \le K \le k_6(\rho)$, one has $N_D < W$.*

Proof. For $\rho \le 1/4$, the only possible new modes beyond the first modes $\ell = 0, \dots, 4$ are $(0,2), (1,2)$; the first $\ell = 5$ mode is delayed by $j_{11/2,1} > 17/2$. For $\rho \ge 1/4$, $z > 4$ makes every second mode absent through k_6, and first modes $\ell = 0, \dots, 5$ are the complete inventory. The cumulative first-mode caps are $1, 4, 9, 16, 25, 36$; the two possible second modes give caps $26, 29$. Their entry walls are

$$\begin{array}{c|c} (1,1) & k_1 \\ (2,1) & \max\{k_2, 51/10\} \\ (3,1) & \max\{k_3, 13/2\} \\ (4,1) & \max\{k_4, 15/2\} \\ (5,1) & \max\{k_5, 17/2\} \\ (0,2) & 2z \\ (1,2) & \max\{p_1, 77/10\}. \end{array}$$

The initial cap is paid uniformly because

$$W(\rho, z_\rho) = \frac{2\pi^2}{9}\frac{1+\rho+\rho^2}{(1-\rho)^2} > \frac{2\pi^2}{9} > 2 > 1.$$

For a moving wall $T^2 = az^2 + b$, $W(\rho, T)$ increases whenever $a\pi^2(1+\rho) > b\rho^2(1-\rho)^2$, which holds here because $b/a \le 42$. Fixed walls decrease with ρ. Exact payments are obtained by the splits

$$(k_1, 1/8, 4), \; (k_2, 3/10, 9), \; (k_3, 1/2, 16), \; (k_4, 1/2, 25), \; (k_5, 13/25, 36), \; (p_1, 1/5, 29).$$

Writing $A = a(333/106)^2/(1-r)^2 + b$ and $B = 7(1-r^3)/99$, each moving payment is the positive rational check $B^2A^3 - C^2 > 0$. In the displayed order the numerators are

$$153864824754340538785, 139981084807345785305 3, 137027505353736631,$$
$$2507119282010251109, 926724046867387424 34741, 373255772037478993002833,$$

over positive denominators. The corresponding fixed-side margins are positive as well; for example they are $1387329/11000000, 6277/6336, 775/704, 452843/343750$, and $849901/281250$ at the nontrivial splits. Finally $W(\rho_c, 2z) > 26$. Every possible mode is therefore paid, including coincident entries, and every equality remains assigned to the lower cap. Parts I, II, and IV of the *Purely Analytic Supplement* map every printed moving numerator and fixed margin to its wall and cap. $\square$

Lemma 4.3 (Lower side through d)**.** *For $\rho_0 < \rho < \rho_c$ and $(2\rho)^{-1} < K \le d$, one has $N_D < W$.*

Proof. Because $3z > d$, every $n \ge 3$ mode is absent. The zero registry and angular propagation exclude first modes $\ell \ge 6$ and second modes $\ell \ge 3$. The complete cap table is

open-lower, closed-upper range	N_D cap
$L < K \le 4$	1
$4 < K \le 51/10$	4
$51/10 < K \le 13/2,\ K \le 2z$	9
$51/10 < K \le 13/2,\ K > 2z$	10
$13/2 < K \le 15/2,\ K \le 2z$	16
$13/2 < K \le 15/2,\ K > 2z$	17
$15/2 < K \le 77/10$	26
$77/10 < K \le 17/2$	29
$17/2 < K \le 9$	40
$9 < K \le d$	45.

In fact $13/2 < 2z < 15/2$, so the cap-10 row is empty. The fixed-wall lower bounds, using $\rho < \rho_c < 7/50$, exceed their caps by

$$\frac{793292}{1546875}, \frac{486236661}{1375000000}, \frac{333100003}{99000000}, \frac{329797}{88000}, \frac{3590476297}{1125000000}, \frac{327078887}{99000000}, \frac{8805519}{1375000},$$

respectively. The initial cap is paid by $W(\rho_c, L(\rho_c)) > 19/6 > 1$, and the moving branch by $W(1/19, 2z) > 508/27 > 17$. This proves every row, including $K = d$. Parts I and IV of the *Purely Analytic Supplement* record each margin with its corresponding row and prove the moving $2z$ payment. $\square$

The upper-floor comparisons are strict. On the lower side, $H_0(\rho_*) = (2\rho_*)^{-1} > d$, while $K_0 > C_0/\omega_0 > 12 > d$. On the high side, $K_0 > 12 > k_6$ for $\rho \le 1/2$, while $K_0 > 384 > k_6$ for $1/2 < \rho < 39/50$. Thus exact subtraction gives

$$\boxed{\begin{aligned} \mathcal{D}_{19} = \{&\rho_* < \rho \le \rho_0 : (2\rho)^{-1} < K < U(\rho)\} \\ &\cup \{\rho_0 < \rho < \rho_c : d < K < U(\rho)\} \\ &\cup \{\rho_c \le \rho < 39/50 : k_6(\rho) < K < U(\rho)\}. \end{aligned}} \tag{4.17}$$

At $\rho = \rho_0$, $(2\rho_0)^{-1} = d$, so the would-be new lower fiber is empty and the first component owns the equality slice.

4.4 Combined closure of the lower components and the high staircase

Set
$$\sigma := \frac{3\omega_0}{4}.$$
We make every comparison in the required ordering explicit. The alternating Machin sums give
$$\pi < 16\left(\frac{1}{5} - \frac{1}{3\cdot 5^3} + \frac{1}{5\cdot 5^5} - \frac{1}{7\cdot 5^7} + \frac{1}{9\cdot 5^9}\right) - 4\left(\frac{1}{239} - \frac{1}{3\cdot 239^3}\right) < \frac{355}{113},$$
and the last gap is exactly
$$\frac{45167474711}{189820334689453125} > 0.$$
Since
$$3\cdot 1101^2\cdot 113^2 - 607^2\cdot 355^2 = 1998482 > 0,$$
we have
$$\sqrt{3} > \frac{607\cdot 355}{1101\cdot 113} > \frac{607\pi}{1101},$$
and therefore
$$\omega_0 = \frac{\sqrt{3}}{2\pi} - \frac{1}{6} > \frac{40}{367}. \tag{4.18}$$
On the other side,
$$25\cdot 333^2 - 243\cdot 106^2 = 41877 > 0$$
and $\pi > 333/106$ imply $9\sqrt{3} < 5\pi$, hence $\omega_0 < 1/9$. Moreover $2\cdot 32^2 - 45^2 = 23 > 0$, so $\sqrt{2} > 45/32$, $C_0 > 7/4$, and
$$\rho_* = \frac{\omega_0}{2C_0} < \frac{2}{63}.$$
Now all remaining comparisons are witnessed by
$$4\cdot 337 < 63^2, \qquad 18^2 < 337, \qquad \frac{1}{18} < \frac{3}{40} < \frac{3\omega_0}{4} < \frac{1}{12} < \frac{7}{51},$$
$$\frac{7}{51} < \frac{1}{1+2\pi} < \frac{1}{7}, \qquad \frac{1}{7} < \frac{39}{50}.$$
Here the first inequality on the second line uses $\pi < 22/7$, and the second uses $\pi > 3$. Thus, without decimal comparison,
$$\rho_* < \rho_0 < \sigma < \rho_c < \frac{39}{50}.$$

Lemma 4.4 (Closure of both lower components of $\mathcal{D}_{19}$)**.** *The strict inequality $N_D < W$ holds on the first two components of* (4.17)*.*

Proof. We divide the proof into a wedge, a finite staircase, and one remaining floor cell.

For a low-interface tail beginning at $x_r = r + 1/2 < \mu = \rho K$, put
$$n_r = \lfloor \mu - x_r \rfloor, \qquad q_r = x_r + n_r, \qquad p = \lfloor \omega_0 K \rfloor.$$
The concave-head/convex-tail split in (2.46), now with its coarse first-shelf payment, gives
$$\frac{\mathcal{T}_r}{2} < \int_{x_r}^{K} A_{\rho,K}(z)\,dz + \delta_r - \frac{p - n_r}{4}, \qquad 0 \le \delta_r < \frac{2(\mu - q_r)^{5/2}}{15\sqrt{\mu}}. \tag{4.19}$$

When $n_r = 0$, the concave head is absent and the formula follows from the convex tail alone. From (4.18) and $400 \cdot 337 - 367^2 = 111 > 0$,

$$\omega_0 d > \frac{40}{367}\frac{\sqrt{337}}{2} = \frac{20\sqrt{337}}{367} > 1.$$

If $0 < \rho \le \sigma$, $K > d$, and $\rho K > 1/2$, then with $x = \omega_0 K > 1$ and $m = \lfloor x \rfloor$,

$$n_r \le \left\lfloor \frac{3x}{4} - \frac{1}{2} \right\rfloor < m = p.$$

Since $\delta_r < 2\sqrt{2}/15 < 1/4$, every tail in (4.19) is strict. The hypotheses hold throughout the first component of $\mathcal{D}_{19}$ and on $\rho_0 \le \rho \le \sigma$, $K > d$. This includes the complete $\rho = \rho_0$ fiber.

For $\rho_0 < \rho < \rho_c$ and $K \le \sqrt{114}$, the complete zero registry and (4.9) give the following exhaustive inventory:

range (lower face open, upper face closed)	newly possible modes	cap
$d < K \le 10,\ K \le 3z$	first 0:5, second 0:2	45
$d < K \le 10,\ K > 3z$	preceding plus $(0,3)$	46
$10 < K \le 81/8$	plus $(6,1)$, possibly $(0,3)$	59
$81/8 < K \le 21/2$	plus $(3,2)$	66
$21/2 < K \le \sqrt{7073}/8$	plus $(1,3)$	69
$\sqrt{7073}/8 < K \le \sqrt{114}$	plus $(4,2)$	78.

First modes $\ell \ge 7$, second modes $\ell \ge 5$, third modes $\ell \ge 2$, and all $n \ge 4$ are absent. In particular,

$$(81/8)^2 + 8 = 7073/64$$

is the propagated $(4,2)$ wall. Since $\rho < \rho_c < 1/7$, the uniform estimate

$$W(\rho, K) > \frac{2}{9(22/7)}\left(1 - \frac{1}{7^3}\right) K^3$$

pays the successive caps with exact positive reserves

$$\frac{2908}{539}, \quad \frac{6199}{539}, \quad \frac{990435}{137984}, \quad \frac{555}{44}, \quad \frac{159}{44}.$$

The first number pays even cap 46 from $K > 9$; the last pays cap 78 from $K > 21/2$. Hence $K = \sqrt{114}$ is included.

It remains to consider $\sigma < \rho < \rho_c$ and $K > \sqrt{114}$. Put

$$p = \lfloor \omega_0 K \rfloor, \qquad m = \lfloor \rho K - 1/2 \rfloor.$$

Then $p \ge 1$. Except on

$$\mathcal{B} = \left\{ \sigma < \rho < \rho_c, \quad K > \sqrt{114}, \quad \rho K \ge 5/2, \quad K < 2/\omega_0 \right\}, \tag{4.20}$$

one has $m \le 2p - 1$. Indeed this is immediate when $\rho K < 5/2$; when $K \ge 2/\omega_0$, one has $p \ge 2$. Moreover,

$$\rho_c < \frac{1}{7} < \frac{60}{367} < \frac{3\omega_0}{2},$$

where the first and last comparisons were proved above and $367 < 420$. Hence $\rho K - 1/2 < 2p$.

Away from a half-integer μ, all interface remainders in (4.19) are equal. Summing over $r = 0, \ldots, m$ gives the average floor reserve

$$\frac{1}{4}\left(p - \frac{m}{2}\right) \ge \frac{1}{8}.$$

For $m = 0$, use $\delta < 1/4$; for $m \geq 1$, $\mu > 3/2$ and $\delta < 2/(15\sqrt{3/2}) < 1/8$. At $\mu \in \mathbb{Z} + 1/2$, assign the interface to the high side; then the remainder is zero and strictness remains in the nonzero convex tail.

On $\mathcal{B}$, $\omega_0 > 40/367$, so $K < 367/20 =: R$. The turning estimate

$$A_{\rho,K}(z) \leq G_K(z) < \frac{(K-z)^{3/2}}{3\sqrt{K}}$$

and monotonicity in K reduce the proxy count at $z_\ell = \ell + 1/2$ to

$$\frac{(357 - 20\ell)^3}{1321200} < \left(c_\ell + \frac{3}{4}\right)^2.$$

The integer numerators of these strict comparisons are

ℓ	0	1	2	3	4	5	6	7	8	9	10	11	12	13	14
c_ℓ	6	5	5	4	4	3	3	3	2	2	1	1	1	1	0

and, in the same order,

$$14697882, 5409422, 11827162, 3611502, 8555642, 1604782, 5267322, 8361062,$$
$$2346202, 4446342, 176282, 1474822, 2444562, 3133502, 286642.$$

The $\ell = 14$ row and monotonicity remove the entire angular tail. Therefore the full multiplicity cap is

$$\sum_{\ell=0}^{13}(2\ell+1)c_\ell = 395.$$

But $\rho K \geq 5/2$, $\rho < \rho_c < 11/80$, and $1/\pi > 113/355$ give

$$W(\rho, K) > \frac{160293325}{378004} = 395 + \frac{10981745}{378004} > 395.$$

Thus $\mathcal{B}$ is closed as well. The face $\rho K = 5/2$ is $\mathcal{B}$-owned and $K = 2/\omega_0$ is aggregate-owned. All lower components of $\mathcal{D}_{19}$ are proved. □

We now turn to the high component. Define the strict channel proxy

$$b_{\ell,n}(z)^2 = \max\{n^2z^2 + \ell(\ell+1),\, \beta_{\ell,n}^2\}, \tag{4.21}$$

For the radial channel we set, for every $n \geq 1$,

$$\beta_{0,n} := j_{1/2,n} = n\pi,$$

the last identity following from $J_{1/2}(x) = \sqrt{2/(\pi x)}\sin x$. For $\ell \geq 1$, the first-mode β's are in (4.13) and

$$\beta_{1,2} = 77/10, \quad \beta_{2,2} = 9, \quad \beta_{3,2} = 103/10, \quad \beta_{4,2} = \sqrt{11409}/10,$$
$$\beta_{5,2} = 129/10, \quad \beta_{1,3} = 21/2, \quad \beta_{2,3} = 61/5.$$

By (4.7) and (4.8), a mode can be counted only if $b_{\ell,n}(z) < K$.

Lemma 4.5 (Closed high staircase through k_{11})**.** *For $\rho_c \leq \rho \leq 39/50$ and $z_\rho \leq K \leq k_{11}(\rho)$, one has $N_D < W$.*

Proof. Lemma 4.2 already proves the claim for $z_\rho \leq K \leq k_6(\rho)$, including the common face. It therefore suffices to treat $k_6(\rho) < K \leq k_{11}(\rho)$. The exhaustive checkpoint inventories are

checkpoint	possible modes
k_7	first 0:6, second 0:1, no $n \geq 3$
k_8	first 0:7, second 0:3, no $n \geq 3$
k_9	first 0:8, second 0:4, no $n \geq 3$
k_{10}	first 0:9, second 0:5, third 0:1, no $n \geq 4$
k_{11}	first 0:10, second 0:4, third 0:1, no $n \geq 4$.

For example, at k_{11} the mode $(5, 2)$ would require both $z^2 < 34$ from the product wall and $z^2 > (129/10)^2 - 132 = 3441/100$ from the ball wall, a contradiction. Also $4z > k_{11}$ at $z \geq \pi + 1/2$. Thus radial indices $1, 2, 3, 4$ and angular indices through 11 are proved above to be exhaustive for the stated checkpoints.

Writing H for the highest first channel and h for the highest second channel, the block cap is $(H + 1)^2 + (h + 1)^2$, with an additional 1 or 4 for third modes. The complete cap tables at k_8 and k_9 are

$k_8 : H\backslash h$	$\varnothing$	0	1	2	3
7	64	65	68	73	80
6	49	50	53	–	–
5	36	37	40	45	52
≤ 4	25	26	29	34	41

$k_9 : H\backslash h$	$\varnothing$	0	1	2	3	4
8	81	–	–	–	–	–
7	64	65	68	73	–	–
6	49	50	53	58	65	74
5	36	37	40	45	52	61
≤ 4	25	26	29	34	41	50.

The dashes follow from incompatible necessary inequalities, not from a numerical sample. At k_{10} the cap lists are

$$100; \quad 81, 97; \quad 64, 80, 89, 100; \quad 74, 69, 78, 85, 89,$$

and at k_{11} they are

$$121; \quad 100, 125; \quad 106, 110; \quad 93.$$

Define the exact finite proxy

$$C_z(K) = \sum_{(\ell,n)} (2\ell + 1)\mathbf{1}_{\{b_{\ell,n}(z)<K\}},$$

where the sum runs over the finite inventory above. Between proxy events it is constant and W increases in K. Every squared event is either constant or affine in z^2. At a moving event $K^2 = sz^2 + d_0$, the derivative of its Weyl payment has the sign of

$$s\pi^2 \frac{1 + \rho}{(1 - \rho)^2} - \rho^2 d_0. \tag{4.22}$$

Thus the finite split registry

$$\frac{1}{5}, \frac{3}{10}, \frac{1}{3}, \frac{3}{8}, \frac{2}{5}, \frac{5}{12}, \frac{3}{7}, \frac{1}{2}, \frac{107}{200}, \frac{8}{15}, \frac{3}{5}, \frac{16}{25}, \frac{13}{20}, \frac{2}{3}, \frac{4}{25}, \frac{1}{4}, \frac{7}{20},$$

together with $z^2 = 16, 68/3, 34$, reduces every cell to exact rational endpoint arithmetic. The smallest representative reserves in the six successive bands are

$$\frac{1387329}{11000000}, \quad \frac{849901}{281250}, \quad \frac{835355}{577368}, \quad \frac{835355}{577368}, \quad \frac{943655}{171072}, \quad \frac{507845}{261954}.$$

Every realizable event is strictly paid, with coincident events charged at their combined multiplicity.

There is one conservative row requiring explicit treatment. Angular propagation gives

$$j_{9/2,2}^2 > \left(\frac{103}{10}\right)^2 + 8 = \frac{11409}{100}.$$

If $(4,2)$ were counted at k_9, this ball wall would force $z^2 > 2409/100$, whereas the product wall forces $z^2 < 70/3$; since $2409/100 - 70/3 = 227/300 > 0$, the printed cap-74 cell is actually empty. If one deliberately retains the weaker conservative cell, counting the mode forces $K > 207/20$ and $\rho < 7/20$. Therefore

$$W(\rho, K) > \frac{52823261673}{704000000} = 74 + \frac{727261673}{704000000} > 74. \tag{4.23}$$

This is the corrected cap-74 payment; it is smaller than the generic k_9 summary reserve and must not be hidden by a claim that all omitted cells have larger reserves.

At every $K = k_m$, the $(m, 1)$ mode is on its product equality wall and is excluded. At every fixed Bessel wall the zero inequality is strict. The event calculation therefore covers all equality and coincidence faces, and proves the lemma. □

The combined finite analytic ledger—covering the zero/sign registry, the two-sided and lower staircases, the lower closure, the high staircase, optical constants, and exact subtraction—can be replayed without floating point. Its core contains 488 substantive rational/radical checks and 65 bookkeeping identities; these are whole-ledger counts, not a count of k_{11}-only cells. It also checks the sign of (4.22) on the named faces. Every proof decision is a cross multiplication or positive-side squaring; decimal displays play no role. The disjoint (611)-row allocation and every hand-checkable witness are printed in Parts I–VI of the *Purely Analytic Supplement*.

4.5 The optical closure and the exact residual $\mathcal{D}_{20}$

Lemma 4.6 (All-frequency optical theorem)**.** *For $39/50 \le \rho < 1$ and $K \ge 0$, $N_D(A_{\rho,1}, K^2) \le W(\rho, K)$, with strict inequality for $K > 0$.*

Proof. Put $\varepsilon = 1 - \rho \le 11/50$, $a = \varepsilon K$, and $c = 1126/625$. We use two inclusive screens meeting at $a = c/\varepsilon$.

For the product screen, (4.7) implies that a counted mode satisfies

$$n^2\pi^2 + \varepsilon^2\ell(\ell + 1) < a^2.$$

There is no mode for $a \le \pi$. For $a > \pi$, write $a/\pi = N + t$, with $N \ge 1$, $0 < t \le 1$, assigning $(N, t) = (m - 1, 1)$ at $a = m\pi$. If

$$M_n = \left\lceil \sqrt{(b_n/\varepsilon)^2 + 1/4} - 1/2 \right\rceil,$$

the exact angular multiplicity is bounded by

$$\mathcal{P}_\varepsilon(a) = \sum_{n=1}^{N} M_n^2.$$

Direct summation of the continuous-minus-discrete product deficit gives

$$\frac{D(a)}{\pi^2} = \frac{N^2}{2} + N\left(t^2 + \frac{1}{6}\right) + \frac{2t^3}{3}. \tag{4.24}$$

For $C_D = 1382/3125$, the difference $D/\pi^2 - C_D(N + t)^2$ increases in N, and at $N = 1$ is minimized at $t = C_D$, where it equals $1822532/91552734375 > 0$. Thus $D > C_D a^2$.

The elementary ceiling bound and quarter-circle integral give

$$\varepsilon^2\mathcal{P}_\varepsilon < S_0 + \varepsilon a^2/4 + \varepsilon^2 a/\pi.$$

On $a \le c/\varepsilon$, using $q = 106/333 > 1/\pi$ and $\varepsilon \le 11/50$, the exact reserve is

$$C_D - \left(\frac{2cq}{3} + \frac{11}{200} + (11/50)^2 q^2\right) = \frac{39569}{2772225000} > 0. \tag{4.25}$$

This proves the low screen, including all radial and angular equality walls.

For the complementary-action screen, use the inverse-action construction proved in Lemma 3.2: let the decreasing shell action have inverse $R_{\mathcal{A}}$, put $F = R_{\mathcal{A}}^2$, and $g = -F'$. If τ is the actual curvature interface and $T = a/\pi$, that lemma gives

$$g \text{ decreasing on } (0,\tau), \quad g \text{ increasing on } (\tau, T),$$

$$F(0) = a^2, \quad F(\tau) = \rho^2 a^2, \quad F(T) = 0, \quad g(T) = 2\pi\rho a.$$

For $C(t) = \#\{n : n - 1/4 < t\}$, set $w = t - C(t)$ and $\Psi(t) = \int_0^t w(s)\,ds - t/4$. The complete sawtooth and Stieltjes calculation in Lemma 3.4, including the ungridded curvature interface τ, gives

$$D_{\text{rad}} := \int_0^T F(t)\,dt - \sum_{n-1/4<T} F(n - 1/4) \ge \frac{\rho^2 a^2}{4} - \frac{\pi\rho a}{4}. \tag{4.26}$$

The exact half-integer angular count from Lemma 3.3 satisfies $\varepsilon^2 M_\varepsilon(x)^2 < (x + \varepsilon/2)^2$, even on an angular wall. Since fewer than $a/\pi + 1/4$ radial layers occur, the angular error is strictly below

$$E = \left(\frac{a}{\pi} + \frac{1}{4}\right)\left(\varepsilon a + \frac{\varepsilon^2}{4}\right). \tag{4.27}$$

On $a \ge c/\varepsilon$, the lower screen from (4.26) decreases and the upper screen from (4.27) increases with ε. At $\varepsilon = 11/50$, their exact difference is

$$\frac{14817541}{472867032960000} > 0. \tag{4.28}$$

Thus the strict layer-cake sum is below the action integral, and the phase bridge proves the high screen. Both screens include $a = c/\varepsilon$; at $K = 0$ both sides vanish. $\square$

Combining Lemmas 4.4, 4.5, and 4.6 with (4.17) leaves exactly

$$\boxed{\mathcal{D}_{20} = \left\{(\rho, K) : \rho_c \le \rho < \frac{39}{50}, \quad k_{11}(\rho) < K < U(\rho) = K_0(\rho)\right\}.} \tag{4.29}$$

Indeed $k_{11} < U$: on $\rho \ge \rho_c$, $a_\rho \ge 1$ and $\eta_\rho < 1/8$, so $K_0 > 64$, whereas $k_{11} < 64$ for $\rho < 39/50$. Moreover $\omega_0 < 1/9 < 7/51 < \rho_c$, so H_0 is not eligible on the surviving ratio range and $U = K_0$. The face $K = k_{11}$ belongs to the closed staircase, $\rho = 39/50$ belongs to the optical theorem, and $K = U$ belongs to the fixed-ratio high-frequency owner.

4.6 All-frequency analytic closure of the exact residual $\mathcal{D}_{20}$

The final residual is closed without a frequency split. The retained- remainder identity and angular-block estimate are global, while the scalar gap is strictly increasing in K above its positive base curve. The detailed calculations are printed in Parts VII–IX of the accompanying *Purely Analytic Supplement*; the logical assembly is included here.

Lemma 4.7 (All-frequency analytic middle theorem)**.** *For*

$$\rho_c \le \rho \le \frac{39}{50}, \qquad K \ge k_{11}(\rho),$$

one has $N_D(A_{\rho,1}, K^2) < W(\rho, K)$.

Proof. Let P_{coarse} be the strict lattice proxy in (4.1). Part VII of the *Purely Analytic Supplement* derives an exact retained-remainder identity and a smooth quantity $\mathcal{H}$ such that

$$\mathcal{E}_{\mathrm{ang}} < \mathcal{H} \quad \Longrightarrow \quad P_{\mathrm{coarse}} < W. \tag{4.30}$$

Here $\mathcal{E}_{\mathrm{ang}}$ is the exact error made when the inverse radial action is sampled on the angular half-integer lattice; the strict counting convention is built into its definition.

On the displayed closed domain,

$$\frac{(1-\rho)k_{11}(\rho)}{\pi} = \sqrt{1 + \frac{132(1-\rho)^2}{\pi^2}} > 1,$$

so at least one complete radial block is present. Applying the one-layer left-block bound across the disjoint complete cells gives the strict global estimate below; its strict rounding convention includes the common radial–angular jump faces.

$$\mathcal{E}_{\mathrm{ang}} < \frac{(1-\rho^2)K^2}{6} - \frac{1}{2}. \tag{4.31}$$

The remaining scalar theorem in Part IX is proved on the base curve $K = k_{11}(\rho)$ by rational secant majorants and positive Bernstein coefficients. Its derivative is strictly positive for every $K > 12$. Since $k_{11}(\rho) > 241/20 > 12$, upward integration gives, for every $K \geq k_{11}(\rho)$,

$$\mathcal{H} - \frac{(1-\rho^2)K^2}{6} + \frac{1}{2} > 0. \tag{4.32}$$

Thus $\mathcal{E}_{\mathrm{ang}} < \mathcal{H}$, and (4.30) yields $P_{\mathrm{coarse}} < W$. The phase reduction (4.1) proves the claim. Every inequality used in (4.31)–(4.32) is derived in Parts VII–IX by integration, exact rational comparison, or positive-side squaring. No estimate has an upper-frequency premise. ☐

Proposition 4.8 (Exact analytic closure of $\mathcal{D}_{20}$)**.** *Every point of the compact-middle residual is covered by Lemma 4.7.*

Proof. The exact residual description (4.29) gives

$$\mathcal{D}_{20} \subset \left\{ (\rho, K) : \rho_c \leq \rho \leq \frac{39}{50}, \quad K \geq k_{11}(\rho) \right\}.$$

The containing set is precisely the closed domain of Lemma 4.7. Hence no residual point remains. The inherited faces $\rho = 39/50$, $K = k_{11}(\rho)$, and $K = U(\rho)$ remain outside $\mathcal{D}_{20}$, while $\rho = \rho_c$ is included subject to its two strict frequency inequalities. The exact owner subtraction is proved row by row in Parts I–VI of the *Purely Analytic Supplement.* ☐

Proposition 4.9 (Compact-middle Pólya theorem)**.** *For every*

$$\rho_* < \rho < \frac{39}{50}, \qquad K \geq 0,$$

the Dirichlet counting function of the concentric spherical shell satisfies

$$N_D(A_{\rho,1}, K^2) \leq W(\rho, K).$$

Proof. Proposition 4.1 proves the inequality outside the exact residual $\mathcal{D}_{16}$ in (4.6). Within that residual, Lemmas 4.2, 4.3, 4.4, and 4.5 remove precisely the regions subtracted in the definition of $\mathcal{D}_{20}$. Hence any point not yet owned belongs to $\mathcal{D}_{20}$ in (4.29). The analytic compact theorem proves the inequality on all of $\mathcal{D}_{20}$ by Proposition 4.8. Thus no residual point remains. The cases on every threshold face have the explicit owners specified above; in particular, equality at a frequency wall is not lost in taking complements. Finally, $K = 0$ is immediate from $N_D(A_{\rho,1}, 0) = W(\rho, 0) = 0$. This proves the assertion for the whole stated parameter range. ☐

4.7 Analytic dependencies and optional cross-checks

No floating-point computation, interval enclosure, or executable truth value is a premise of the proof. The finite portion consists of exact identities, rational cross-multiplications, positive-side squarings, and explicit sign decompositions. Parts I–VI of the *Purely Analytic Supplement* expose the former 611 ledger predicates in the following normalized disjoint allocation, after each repeated identity is assigned to one canonical owner:

analytic packet	uniquely owned rows
remaining threshold, zero, lower, and cap–74 rows	103
universal arithmetic and checkpoint inventories	278
connected high-event registry	83
seam and two-sided payments	57
upper owner and optical screens	22
exact owner subtraction	68
total	611.

The normalized packet census checks that all pairwise intersections are empty and that the union is the full ledger. Thus a repeated identity is assigned exactly once, not counted once per place where it is used.

Parts VII–IX prove the all-frequency analytic middle theorem by the exact retained-remainder identity, the paired angular-block inequality, and an upward-monotone scalar gap. Together with Parts I–VI, these arguments replace the former compact and aggregate certificates and the executable state-classifier in the logical dependency graph. The former aggregate curvature argument is retained separately only as an optional independent cross-check; it is not a premise of the theorem.

The earlier exact-arithmetic and interval programs are retained separately as optional regression tools. Agreement with them is useful evidence against transcription errors, but their outputs, precision settings, and hashes are not mathematical assumptions and are not used in any implication above.

5 Proof of the spectral theorem

We now assemble the regional results without importing any new estimate. First, elementary bounds give

$$3\sqrt{3} > 5 > \pi,$$

and hence $\omega_0 > 0$. Also $\sqrt{3} < 2$ and $\pi > 3$ give $\omega_0 < 1/6$. Since $C_* > 0$,

$$0 < \rho_* < \omega_0 < \frac{1}{6} < \frac{39}{50} < 1. \tag{5.1}$$

Therefore

$$(0,1) = (0,\rho_*] \,\dot{\cup}\, (\rho_*, 39/50) \,\dot{\cup}\, [39/50, 1). \tag{5.2}$$

Fix $0 < \rho < 1$. At $K = 0$, positivity in Proposition 2.1 gives

$$N_D(A_{\rho,1}, 0) = 0 = W(\rho, 0). \tag{5.3}$$

Let $K > 0$. If $0 < \rho \le \rho_*$, apply Theorem 2.9. If $\rho_* < \rho < 39/50$, apply Proposition 4.9. If $39/50 \le \rho < 1$, apply Lemma 4.6. The three cases in (5.2) are disjoint and exhaustive, so

$$N_D(A_{\rho,1}, K^2) \le \frac{2}{9\pi}(1-\rho^3)K^3 \qquad (K \ge 0). \tag{5.4}$$

Now

$$\omega_3 = \frac{4\pi}{3}, \qquad L_3 = \frac{\omega_3}{(2\pi)^3} = \frac{1}{6\pi^2}, \qquad |A_{\rho,1}| = \frac{4\pi}{3}(1-\rho^3),$$

so

$$L_3|A_{\rho,1}| = \frac{2}{9\pi}(1-\rho^3). \tag{5.5}$$

Taking $K = \sqrt{\Lambda}$ in (5.4) proves the unit-shell form of Theorem 1.1 for every $\Lambda \geq 0$.

For arbitrary $0 < r < R$, put $\rho = r/R$. The unitary dilation

$$(D_R u)(x) = R^{-3/2} u(x/R)$$

maps $L^2(A_{\rho,1})$ onto $L^2(A_{r,R})$, and its Dirichlet form scales by R^{-2}. Hence

$$\lambda_j^D(A_{r,R}) = R^{-2}\lambda_j^D(A_{\rho,1}),$$

and the strict threshold transforms as

$$N_D(A_{r,R},\Lambda) = N_D(A_{\rho,1}, R^2\Lambda). \tag{5.6}$$

Applying the unit-shell result at $R^2\Lambda$,

$$\begin{aligned} N_D(A_{r,R},\Lambda) &\leq \frac{2}{9\pi}(1-\rho^3)(R^2\Lambda)^{3/2} \\ &= \frac{2}{9\pi}(R^3 - r^3)\Lambda^{3/2} \\ &= L_3|A_{r,R}|\Lambda^{3/2}. \end{aligned}$$

This proves Theorem 1.1 for every $R > 0$, including $R < 1$.

6 Proof that spherical shells do not tile

We prove Theorem 1.2 independently of the spectral argument. Set $\alpha = r/R \in (0,1)$ and scale by $1/R$. It suffices to consider

$$T = A_{\alpha,1} = \{x : \alpha < |x| < 1\}.$$

Every rigid motion has the form $x \mapsto Qx + a$, with Q orthogonal. Because T is radial, $Q(T) = T$, so every congruent copy is a translate

$$T_a = a + T = \{x : \alpha < |x - a| < 1\}.$$

Assume for contradiction that $\{T_a : a \in \mathfrak{C}\}$ has pairwise-disjoint interiors and covers almost everywhere. Duplicate centers are impossible because they give the same nonempty open tile.

Fix a unit vector e and put

$$c = \frac{1+\alpha}{2}e, \qquad \delta = \frac{1-\alpha}{2}.$$

Write $B(x,s) = \{y \in \mathbb{R}^3 : |y - x| < s\}$ for the open Euclidean ball. Then $B(c,\delta) \subset T$. Therefore T_a contains $B(a + c, \delta)$, and disjointness implies

$$|a - b| \geq 1 - \alpha \qquad (a \neq b). \tag{6.1}$$

Thus the centers are locally finite: a bounded set cannot contain infinitely many points with fixed positive separation. They are also countable, since

$$\mathfrak{C} = \bigcup_{m\geq 1} (\mathfrak{C} \cap \overline{B}(0,m))$$

is a countable union of finite sets.

Each tile boundary is a union of two spheres and is three-dimensional null. The countable union

$$Z = \bigcup_{a\in\mathfrak{C}} \partial T_a$$

is therefore null. Exact open-shell coverage is already a special case of almost-everywhere coverage. If closed shells cover exactly or almost everywhere, removing Z shows that their open interiors still cover almost everywhere. Hence it is enough to assume

$$\left|\mathbb{R}^3 \setminus \bigcup_{a\in\mathfrak{C}} T_a\right| = 0. \tag{6.2}$$

Choose one tile T_{a_0} and its outer sphere

$$\Sigma = \{x : |x - a_0| = 1\}.$$

Fix $p \in \Sigma$ and put $u = p - a_0$. For $n \geq 1$, let

$$U_n = B\left(p + \frac{2}{n}u, \frac{1}{n}\right).$$

Every U_n lies outside T_{a_0}, is open, and has positive volume. By (6.2), choose $x_n \in U_n \cap T_{a_n}$ with $a_n \neq a_0$. Since $x_n \to p$ and $|x_n - a_n| < 1$, the centers a_n eventually lie in a fixed bounded set. Local finiteness lets us pass to a subsequence with one fixed center $a \neq a_0$. Thus $p \in \overline{T_a}$.

The point p is not in T_a. Otherwise some $B(p,\varepsilon)$ lies in T_a. Choose

$$0 < t < \min\{\varepsilon, 1 - \alpha\}.$$

Then $p - tu \in T_{a_0} \cap T_a$, contradicting disjoint interiors. Therefore every $p \in \Sigma$ belongs to the boundary of a neighboring tile.

If $p \in \Sigma \cap \partial T_a$, then $|p - a| \in \{\alpha, 1\}$, hence $|a - a_0| \leq 2$. Only finitely many centers satisfy this bound. Consequently Σ is covered by finitely many neighboring inner or outer boundary spheres.

Translate so that $a_0 = 0$. The intersection of Σ with a neighboring sphere $\{x : |x-a| = s\}$, $s \in \{\alpha, 1\}$, satisfies

$$|x|^2 = 1, \qquad |x - a|^2 = s^2,$$

and hence

$$2a \cdot x = |a|^2 + 1 - s^2. \tag{6.3}$$

Because $a \neq 0$, (6.3) is a genuine plane. Its intersection with Σ is empty, a tangent point, or a circle, and therefore has zero surface measure on Σ. A finite union of such sets cannot cover Σ, whose area is 4π.

There is no coincident-sphere exception. A neighboring outer sphere equals Σ only when it has the same center, which is the original tile or a forbidden duplicate. A neighboring inner sphere cannot equal Σ because $\alpha < 1$. We have reached a contradiction. Scaling back proves Theorem 1.2.

A External inputs and verification boundary

The proof above is self-contained with respect to every shell-specific spectral reduction, weighted count, endpoint estimate, regional subtraction, fixed-channel variational comparison, seam assignment, and finite exact-rational predicate. It uses the following conventional external results, each only in the scope stated where it enters the proof:

1. standard spherical-harmonic decomposition, regular Sturm–Liouville theory, and the classical Bessel identities collected in [5];
2. the uniform one-variable Bessel-phase enclosure of [2], together with the explicitly quoted floor-sum and turning-point inequalities of [1, 3];
3. Lorch's strict lower bound for the first positive zero of J_ν, in precisely the half-integer specializations derived in the text [4].

All shell-specific finite decisions are exposed in the accompanying *Purely Analytic Supplement*. They require only exact integer and rational arithmetic, elementary integral inequalities, and the classical real functions already present in the analytic argument. The archived interval programs have no place in this verification boundary; they are optional regression checks only.

References

[1] N. Filonov, M. Levitin, I. Polterovich, and D. A. Sher, Pólya's conjecture for Euclidean balls, *Invent. Math.* **234** (2023), 129–169. doi:10.1007/s00222-023-01198-1.

[2] N. Filonov, M. Levitin, I. Polterovich, and D. A. Sher, Uniform enclosures for the phase and zeros of Bessel functions and their derivatives, *SIAM J. Math. Anal.* **56** (2024), no. 6, 7644–7682. doi:10.1137/24M1642032.

[3] N. Filonov, M. Levitin, I. Polterovich, and D. A. Sher, Pólya's conjecture for Dirichlet eigenvalues of annuli, *J. Lond. Math. Soc.* **113** (2026), no. 2, e70425. doi:10.1112/jlms.70425.

[4] L. Lorch, Some inequalities for the first positive zeros of Bessel functions, *SIAM J. Math. Anal.* **24** (1993), no. 3, 814–823. doi:10.1137/0524050.

[5] NIST Digital Library of Mathematical Functions, Chapter 10: Bessel functions, `https://dlmf.nist.gov/`, accessed 15 July 2026.